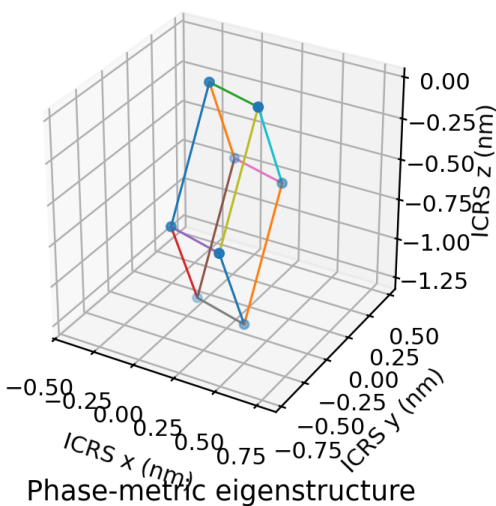


K-FIELD AS A PROJECTIVE EXTERIOR LATTICE

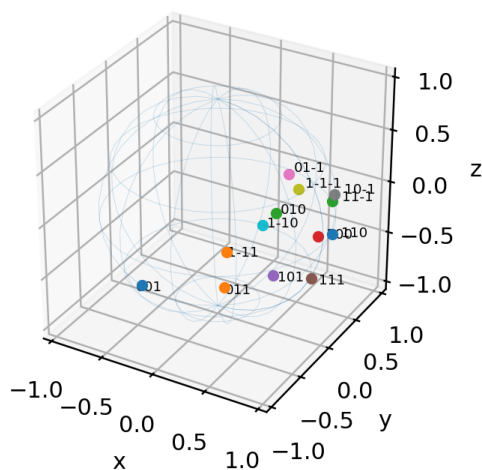
Hidden geometric derivatives, reciprocal-shell closure, scale algebra, phase registry, and eigen-like dimensions

K-Field as a Projective Exterior Lattice

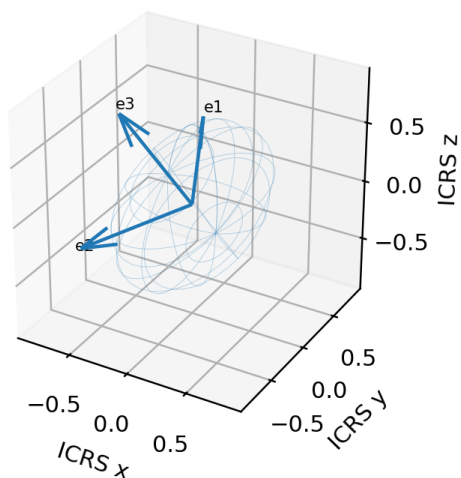
Authoritative Cell3



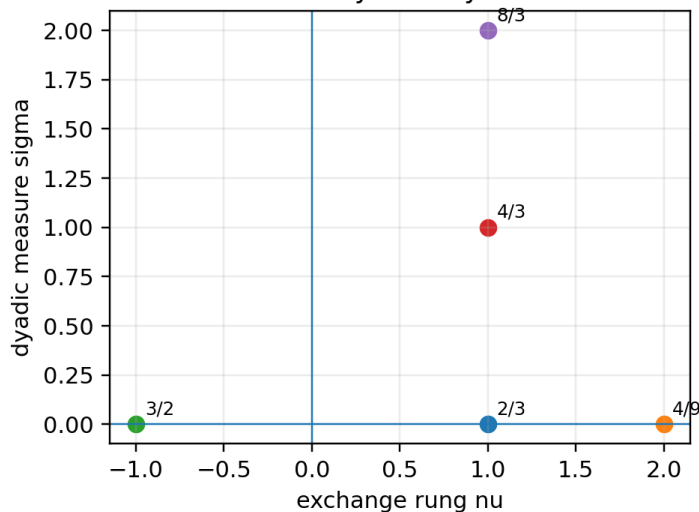
First reciprocal projective shell



Phase-metric eigenstructure



Rank-two binary-ternary scale lattice



Title figure. Four actual-data views of the same architecture: authoritative Cell3 geometry, the complete first reciprocal projective shell, the phase-metric eigensystem, and the rank-two binary-ternary scale lattice.

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Abstract

The K-Field is a fixed, oriented, anisotropic geometric architecture beneath the derivative languages of known science. Its local root is the non-orthogonal Cell3 basis; its collective root is the $3 \times 3 \times 3$ Lucas Group; its internal hierarchy is generated by exact division by 3 and repeated dyadic subdivision; and its observable renderings are carried by direct geometry, reciprocal geometry, exterior grade, phase registry, polarity, occupation, transport, and observer projection. This paper develops the hidden structure that becomes visible when angular, ratiometric, orbital, material, and galaxy-plane results are treated as different projections of one address system.

The central result is that the K-Field is not adequately described as a rectilinear lattice projected onto a unit sphere. It is a projective exterior lattice with a rank-two scale algebra and an attached phase fiber. Direct occupancy directions are generated by the direct metric G ; plane and gradient directions are generated by the reciprocal metric G^{-1} ; reciprocal-plane intersections close exactly onto direct integer rays; the thirteen MOD radial units form the complete nonzero first ternary shell modulo antipodes; the Mass Bridge is the axial sub-spectrum of that shell; and the full shell reconstructs the normalized reciprocal metric. The solar-system rotation, mass, and orbital ledgers then occupy low-dimensional closure manifolds determined by known mechanics, while their residual states retain K-Field phase information.

Foundational determination. Integer occupation, direct/reciprocal metric duality, projective angular rendering, rank-two scale address, sheet and A2 coordinates, exterior grade, phase registry, polarity, and occupation amplitude form the minimum state architecture sufficient to reproduce the recurring structure across the complete corpus. Known physical laws remain unchanged and act as the derivative closure maps that transport the deeper addresses into measured observables.

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1. Scope, definitions, and coordinate contract

This work is a standalone mathematical and physical reference. It begins with the authoritative Cell3 numerical basis and derives every later object from that basis or from the explicitly listed observational ledgers. A geometric identity is kept distinct from a derivative physical interpretation, and a derivative interpretation is kept distinct from an empirical witness. This ordering makes the framework internally reproducible: geometry first, address rule second, physical closure third, and observation fourth.

1.1 ICRS coordinate system

All three-dimensional vectors are expressed in the International Celestial Reference System Cartesian frame. The x axis points toward right ascension 0 deg and declination 0 deg; the y axis points toward right ascension 90 deg and declination 0 deg; and the z axis points toward declination +90 deg. A unit direction with right ascension RA and declination Dec is

$$n(RA, Dec) = \begin{bmatrix} \cos(Dec) \cos(RA), \\ \cos(Dec) \sin(RA), \\ \sin(Dec) \end{bmatrix}^T$$

The unit sphere identifies direction but not distance. Plane normals are projective: a normal and its negative describe the same unoriented plane. Accordingly, angular plane data live naturally in the real projective plane rather than on an oriented sphere.

1.2 Direct, reciprocal, and projective objects

The direct basis B maps integer or fractional native addresses into physical displacement. The reciprocal basis B^{-T} maps integer plane addresses into physical covectors and plane normals. Normalization maps either class into phase space. Direct and reciprocal states may therefore occupy the same plotted angular coordinates while retaining different geometric grades and different physical meanings.

```
direct displacement:      x_m = B m
reciprocal plane normal: g_h = B^-T h
projective direct state:  xhat_m = x_m / ||x_m||
projective reciprocal state: ghat_h = g_h / ||g_h||
```

The notation [z] denotes an integer ray modulo nonzero integer scaling and antipodal reversal. It records a projective address rather than one chosen numerator.

2. Authoritative Cell3 geometry

Cell3 is the fixed local similarity class from which the direct and reciprocal geometries are derived. Its ordered basis is non-orthogonal and left-handed in the chosen ICRS ordering. Common scaling changes every dimensioned measure but leaves all ratios, normalized metrics, projected shapes, handedness, and angular relations unchanged.

Authoritative Cell3: all vertices, edges, faces, and ordered basis

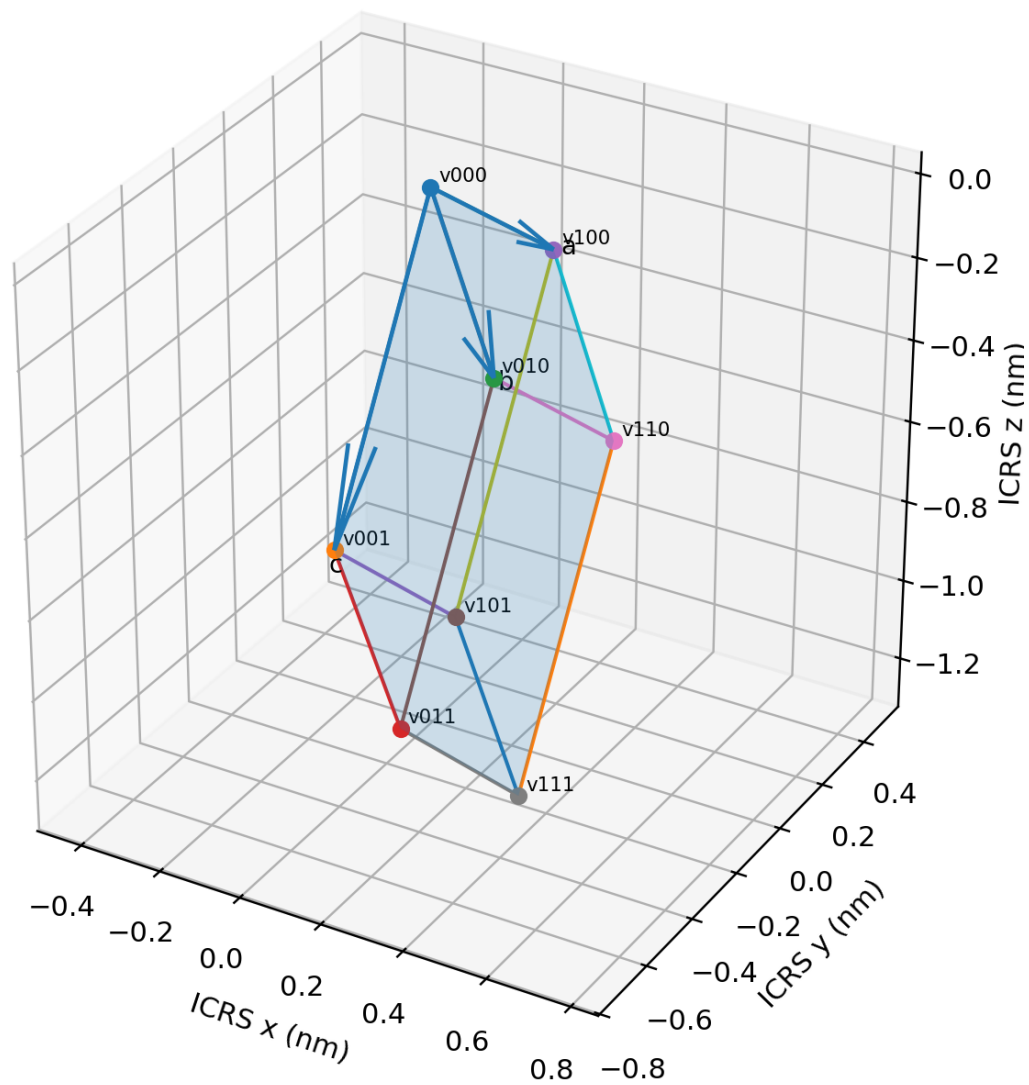


Figure 2. Authoritative Cell3 generated directly from the ordered basis. All eight vertices, twelve edges, six faces, and three primitive vectors are shown with equal physical axis scaling.

2.1 Ordered basis and dimensional invariants

Vector	x (nm)	y (nm)	z (nm)	Length (nm)
a	0.327501242069651	-0.054897007886313	-0.037487502896305	0.334179679073151
b	0.035330775844051	0.207319850441498	-0.547401085391957	0.586410890414836
c	-0.010129317626665	-0.389787702363804	-0.689734408362922	0.792319765037693

Quantity	Value	Unit / meaning
theta_ab	83.933483012312735	deg
theta_ac	80.448130018337906	deg
theta_bc	50.363231813756187	deg
A_ab	0.194869165469900	nm^2
A_ac	0.261106235799531	nm^2
A_bc	0.357809539128562	nm^2
det(B)	-0.117914891916699	nm^3; negative ordered handedness
det(B)	0.117914891916699	nm^3 physical volume

2.2 Complete vertex ledger

Vertex	Native address	x (nm)	y (nm)	z (nm)	Norm (nm)
v000	[0, 0, 0]	0.000000000000	0.000000000000	0.000000000000	0.000000000000
v001	[0, 0, 1]	-0.010129317627	-0.389787702364	-0.689734408363	0.792319765038
v010	[0, 1, 0]	0.035330775844	0.207319850441	-0.547401085392	0.586410890415
v011	[0, 1, 1]	0.025201458217	-0.182467851922	-1.237135493755	1.250773304956
v100	[1, 0, 0]	0.327501242070	-0.054897007886	-0.037487502896	0.334179679073
v101	[1, 0, 1]	0.317371924443	-0.444684710250	-0.727221911259	0.909571953267
v110	[1, 1, 0]	0.362832017914	0.152422842555	-0.584888588288	0.704964152894
v111	[1, 1, 1]	0.352702700287	-0.237364859809	-1.274622996651	1.343653621680

2.3 Similarity invariance

For every positive scale s , the basis sB represents the same Cell3 shape address. Lengths scale as s , areas as s^2 , and volume as s^3 . The normalized direct metric, normalized reciprocal metric, face-area ratios, edge-length ratios, included angles, projective shells, and log-shape point remain fixed.

$B \rightarrow s B$
length $\rightarrow s$ length
area $\rightarrow s^2$ area
volume $\rightarrow s^3$ volume
normalized G and normalized G^{-1} remain invariant

3. Direct and reciprocal metric duality

3.1 Direct metric

The direct Gram matrix $G = B^T B$ contains every intrinsic edge length and included angle. A native displacement $m/2^n$ has squared physical length $m^T G m / 4^n$. The integer graph specifies which states are adjacent; the metric assigns physical distance and angle to those graph transitions.

	a	b	c
a	0.111676057905	0.020710333282	0.043937235098
b	0.020710333282	0.343877732397	0.296392758962
c	0.043937235098	0.296392758962	0.627770610069

3.2 Reciprocal metric and plane families

The rows of B^{-1} evaluate primitive coordinates. A constant primitive coordinate is a physical plane. The reciprocal Gram matrix $R = G^{-1}$ therefore governs plane spacing, gradients, and angular resolution. Direct and reciprocal geometry are not independent structures; they are dual renderings of one Cell3 state.

	a^*	b^*	c^*
a^*	9.208025528684	0.001534802428	-0.645188035677
b^*	0.001534802428	4.903398312306	-2.315175584416
c^*	-0.645188035677	-2.315175584416	2.731171274746

For reciprocal family i , the unit normal is the normalized i -th row of B^{-1} , and the perpendicular spacing associated with one unit change of primitive coordinate is the reciprocal of that row norm.

Reciprocal Cell3 plane families with all 27 Lucas half-step nodes

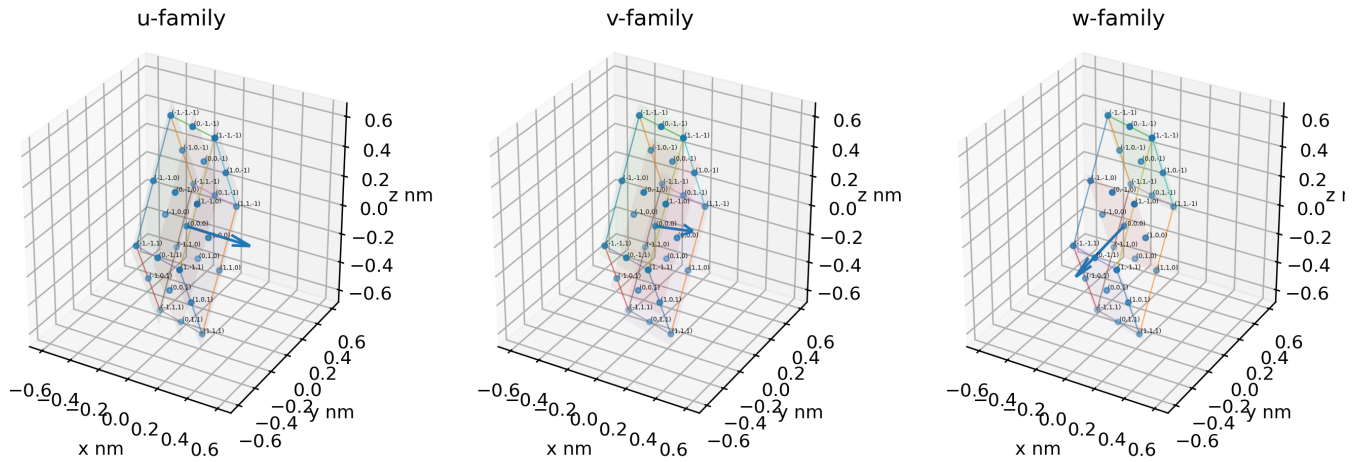


Figure 3. Three reciprocal Cell3 plane families at native offsets -1/2, 0, and +1/2, with all 27 Lucas half-step nodes labelled. The geometry is generated from B and B⁻¹; axes are in nanometres.

Family	Unit normal x	Unit normal y	Unit normal z	Cell3 spacing (nm)	Lucas span (nm)
u / constant a-coordinate	0.995965190309	-0.083602162507	0.032619290564	0.329546529709	0.988639589128
v / constant b-coordinate	0.089052594102	0.866576769291	-0.491033948326	0.451597379724	1.354792139172
w / constant c-coordinate	-0.194092200808	-0.913177168821	-0.358379234791	0.605097741515	1.815293224544

3.3 Recursive plane refinement

At dyadic depth n , each family contains $3 \times 2^n + 1$ planes across one Lucas Group. The total group span remains fixed, while adjacent plane spacing is divided by 2^n . Plane recursion is therefore exactly dual to cell recursion.

$$t_j = -3/2 + j/2^n, \quad j = 0, \dots, 3 \cdot 2^n$$

$$\text{planes per family} = 3 \cdot 2^n + 1$$

$$\text{adjacent physical spacing} = \Delta_i / 2^n$$

4. Projective rendering of rectilinear occupancy

4.1 Direct integer rays on the unit sphere

Let m be a nonzero integer direct address. Its physical displacement is Bm . Normalization removes radial scale and retains only the angular state. The result is an exact map from integer rectilinear occupancy into a discrete projective angular spectrum.

$$\hat{x}_m = Bm / \sqrt{m^T G m}$$

The angular separation of two direct addresses is determined entirely by the integer numerators and the fixed direct metric:

$$\cos(\theta_{mn}) = (m^T G n) / \sqrt{(m^T G m)(n^T G n)}$$

Integer multiples of one address lie on the same projective ray. Angular quantization therefore reflects primitive integer direction classes, not a preferred physical radius.

4.2 Reciprocal projective states

A reciprocal plane address h produces physical normal $B^T h$. Its normalized direction is governed by the inverse metric:

$$\hat{g}_h = B^T h / \sqrt{h^T G^{-1} h}$$

$$\cos(\phi_{hk}) = (h^T G^{-1} k) / \sqrt{(h^T G^{-1} h)(k^T G^{-1} k)}$$

Projective first-shell angular spectra: direct and reciprocal grades

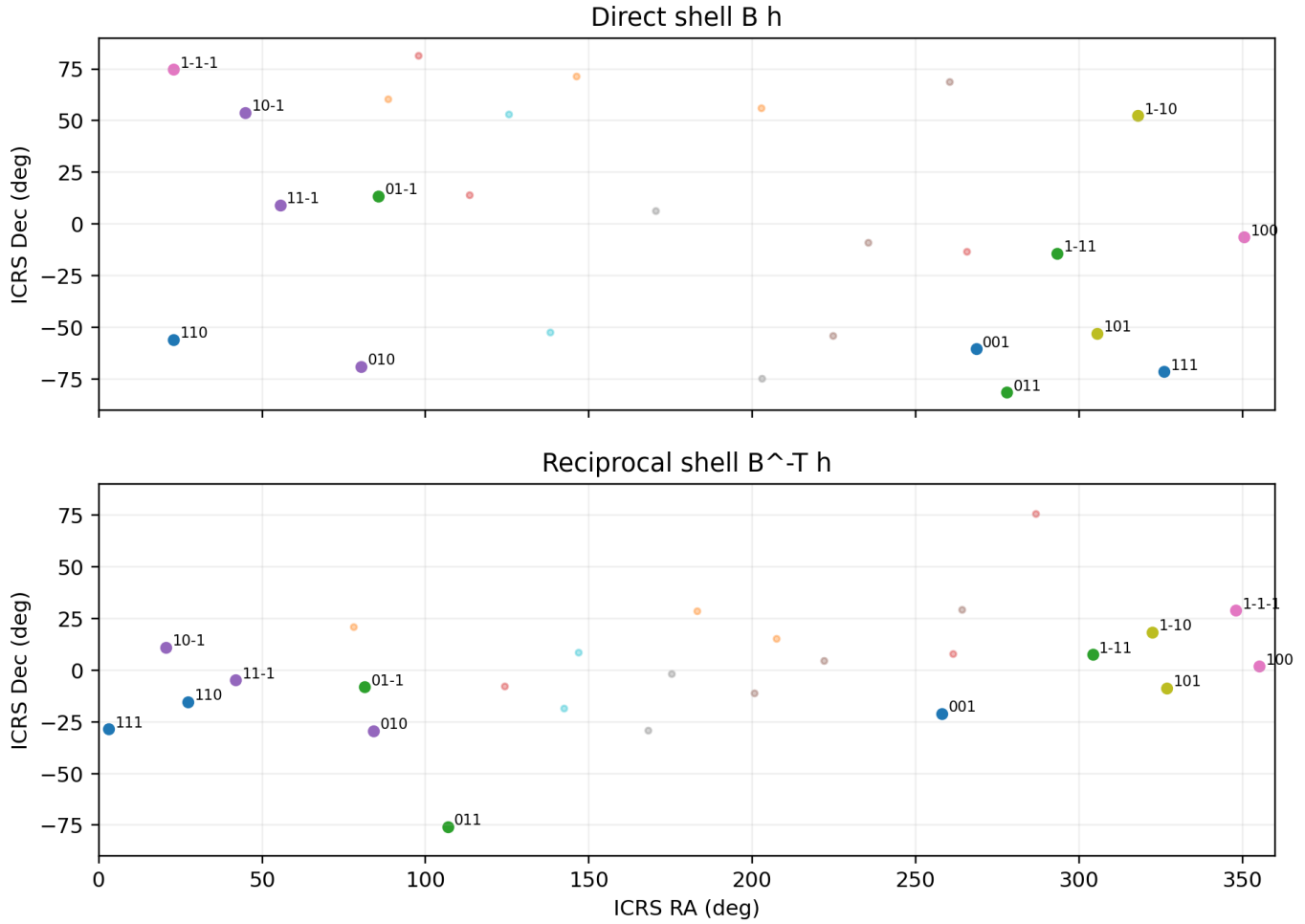


Figure 4. Direct and reciprocal first-shell projective spectra in the fixed ICRS frame. Each labelled point is an integer address; faint points show antipodes.

The two spectra are related by metric duality but are not identical. A body axis, transport direction, or direct occupation ray belongs to the direct grade. A plane normal, gradient, or reciprocal response belongs to the reciprocal grade. A sky catalog that mixes the two without a grade label will display multiple angular families even when every state derives from one Cell3.

5. Projection of three-dimensional plane families into astronomical phase space

5.1 Great circles and small circles

A reciprocal plane has physical equation $\mathbf{g}_h \cdot \mathbf{x} = t$. On a sphere of radius r , $\mathbf{x} = r \mathbf{n}$, so its angular image satisfies

$$\mathbf{g}_{\text{hat}}_h \cdot \mathbf{n} = t / (r \|\mathbf{g}_h\|).$$

When $t = 0$, the projected locus is a great circle. When t is nonzero, the projected locus is a small circle. The angular offset β from the central great circle is

$$\beta = \arcsin[t / (r \|\mathbf{g}_h\|)].$$

At astrophysical distance the argument is small, and β is proportional to $1/r$. Parallel members of one finite-offset rectilinear family therefore become progressively indistinguishable in angular phase space. The physical planes remain distinct in three dimensions even while their sky projections converge.

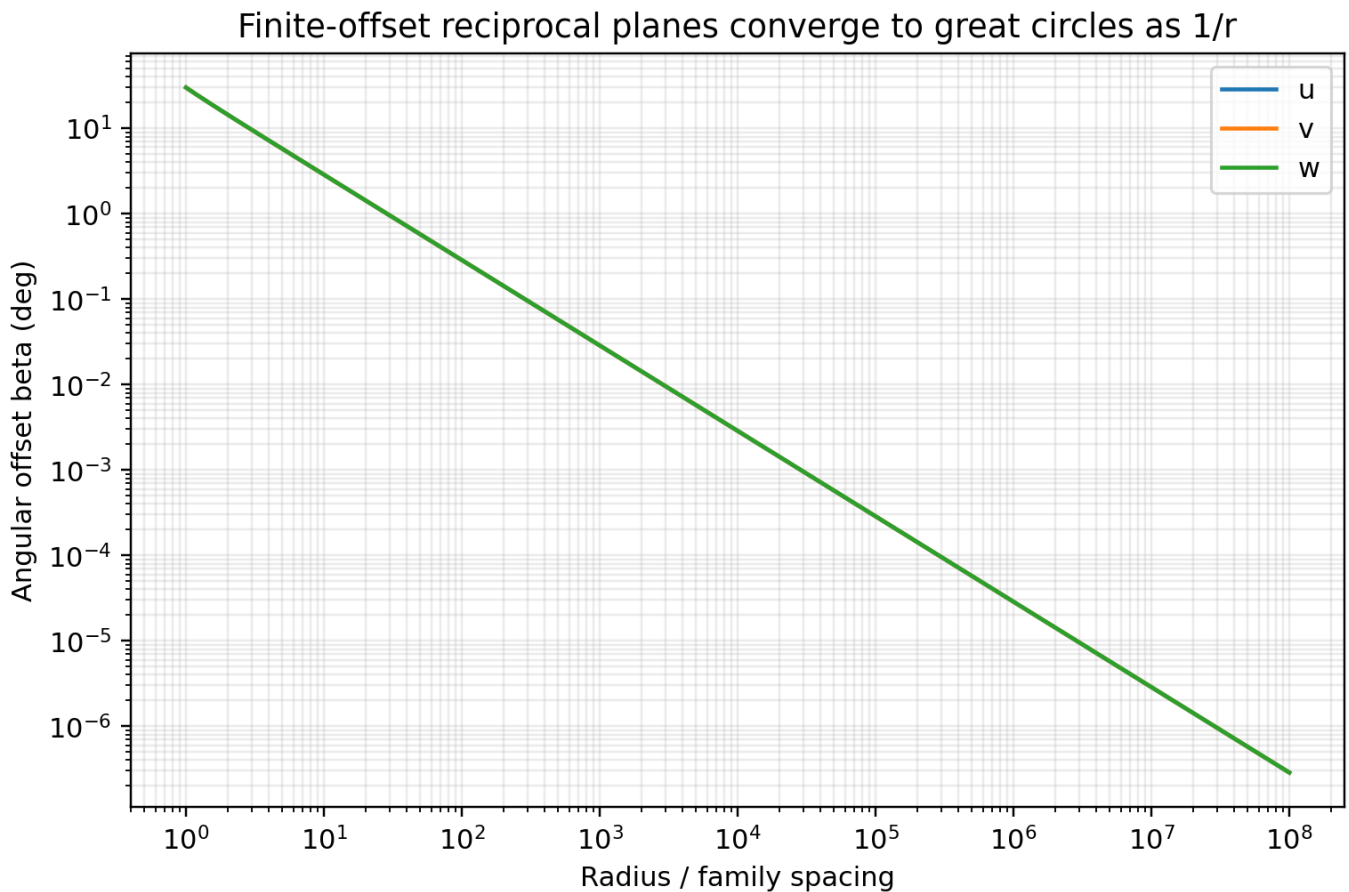


Figure 5. Angular convergence for a half-spacing offset in each authoritative reciprocal family. The independent family spacing cancels under radius normalization, leaving the universal $1/r$ asymptote.

5.2 Observable consequences

This projection law explains why one underlying rectilinear family can appear as a narrow great-circle band, why several parallel structures can converge toward one projective direction, and why physical depth is not recoverable from angular alignment alone. Distance restores the hidden offset coordinate. A valid three-dimensional reconstruction must fit both the great-circle family and the predicted $1/r$ broadening or displacement.

6. Exterior closure: reciprocal-plane intersections are direct-lattice rays

The three-dimensional exterior-product identity supplies an exact closure between reciprocal and direct grades. For any invertible linear map A ,

$$(A \mathbf{u}) \times (A \mathbf{v}) = \det(A) A^{-T} (\mathbf{u} \times \mathbf{v}).$$

Set $A = B^{-T}$. Then the intersection direction of two reciprocal planes is

$$(B^{-T} \mathbf{h}) \times (B^{-T} \mathbf{k}) = \det(B)^{-1} B (\mathbf{h} \times \mathbf{k}).$$

After normalization, the projective crossing direction is

$$\text{chat}_{hk} = \pm B(\mathbf{h} \times \mathbf{k}) / \sqrt{(\mathbf{h} \times \mathbf{k})^T G (\mathbf{h} \times \mathbf{k})}.$$

The cross product $\mathbf{h} \times \mathbf{k}$ is another integer vector. Consequently, the intersection of two reciprocal-lattice planes is exactly a direct-lattice direction. The converse follows from the same identity: the cross product of two direct rays produces a reciprocal-plane normal.

Reciprocal-plane intersections return to direct-lattice rays

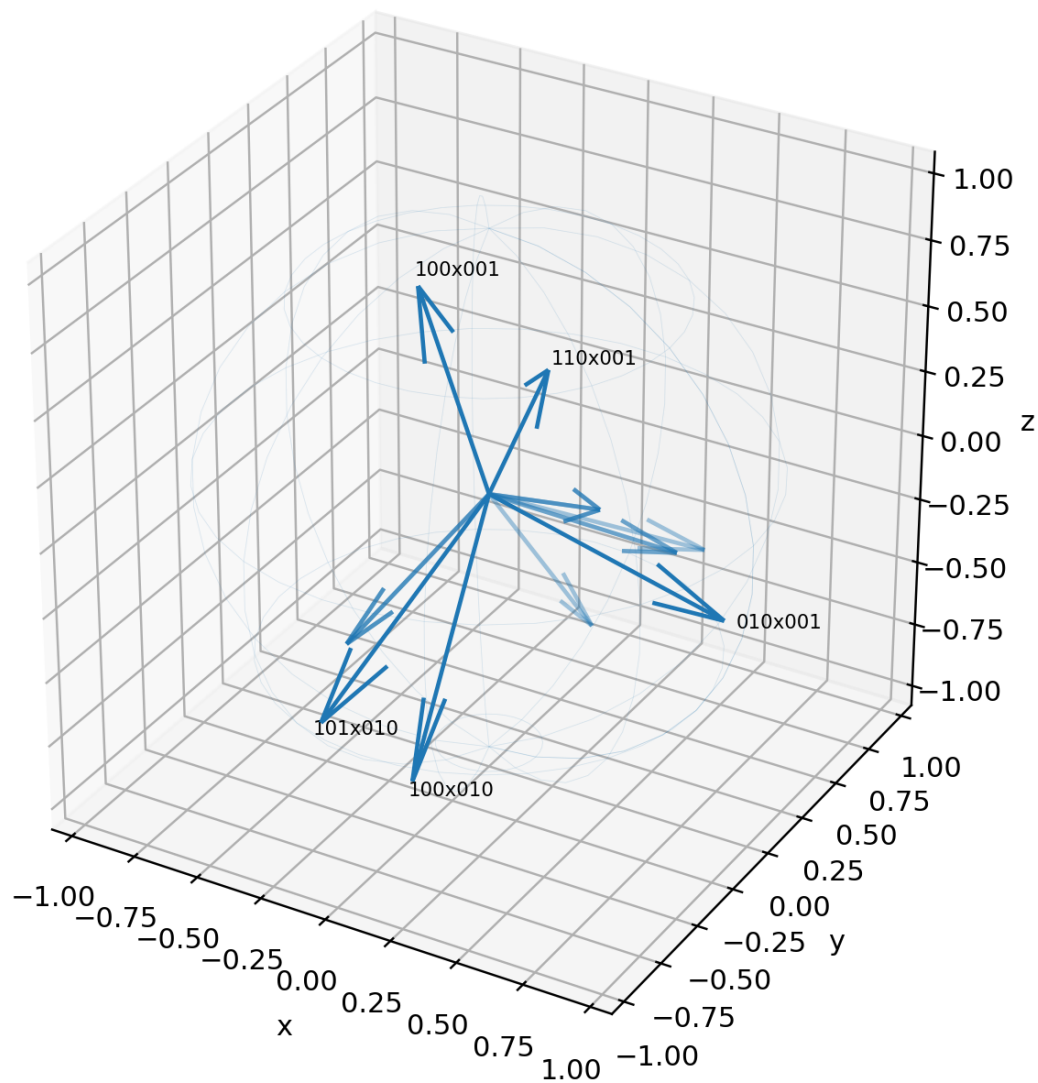


Figure 6. Actual first-shell examples of reciprocal normals and their direct intersection rays. Labels identify the reciprocal address pair; the resulting ray is $B(h \times k)$.

Geometric closure. Direct occupancy, reciprocal planes, and projective crossings are not separate witness classes. They form one exterior algebra in which vectors and bivectors are dual. Galaxy-plane crossing concentrations are therefore reciprocal renderings of direct integer rays.

7. Crossing multiplicity as an arithmetic degeneracy dimension

A projective crossing direction can be represented by many distinct reciprocal-plane pairs. For an integer direct ray $[m]$, define a bounded representation count

$D_H([m]) = \text{number of pairs } (h,k) \text{ with } ||h||, ||k|| \leq H$
and $h \times k$ parallel to m .

This quantity is an eigenvalue-degeneracy-like coordinate. It counts the number of reciprocal exterior products that collapse onto the same direct projective state. High plane-crossing multiplicity is therefore predicted at low-complexity direct rays having large arithmetic representation degeneracy.

In a galaxy-plane analysis, the correct multiplicity statistic counts distinct participating planes rather than repeated pairwise points. That distinction preserves the underlying incidence architecture. A high-multiplicity aperture indicates that many independent reciprocal states share one approximate direct crossing address.

7.1 Back-projection test

For an observed crossing unit vector c , the primitive direct coordinate is proportional to $B^{-1} c$. The hidden-address test is to identify a low-complexity integer vector m parallel to that coordinate. The angular residual is then evaluated in the direct metric rather than ordinary coordinate space.

$m_{\text{candidate}} = \text{primitive integer ray nearest to } B^{-1} c$
 $\text{residual angle} = \arccos[(m^T G B^{-1} c) / \sqrt{(m^T G m)(c^T c)}]$

The resulting address m , representation degeneracy D_H , participating plane families, and phase-cost state form a complete descriptor of each high-order crossing.

8. The complete thirteen-state reciprocal projective shell

The established MOD radial codes are exactly the nonzero ternary address vectors h in $\{-1, 0, +1\}^3$ after antipodal identification. There are $(3^3 - 1)/2 = 13$ projective classes: three axial states, six face-diagonal states, and four body-diagonal states.

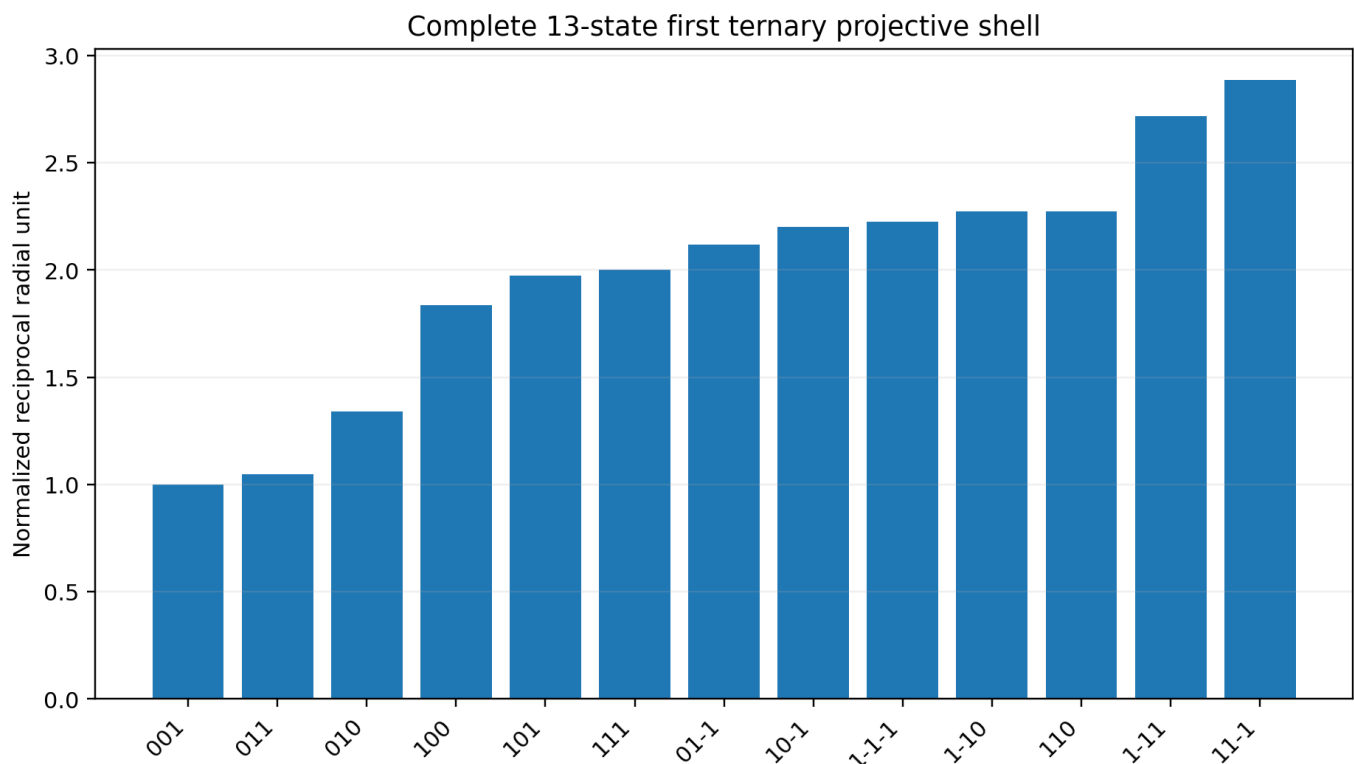


Figure 7. The thirteen reported MOD radial units ordered by the established code sequence. Every value is the reciprocal norm $\sqrt{h^T G^{-1} h}$ normalized to code 001.

Code	Integer h	Class	Reported s_h	Computed s_h	Difference	Reciprocal RA	Reciprocal Dec
001	[0, 0, 1]	axial	1.000000000000	1.000000000000	0.000e+00	258.000576	-21.000693
011	[0, 1, 1]	face diagonal	1.048796632266	1.048796632255	-1.145e-11	106.780627	-75.703940
010	[0, 1, 0]	axial	1.339905341993	1.339905341976	-1.696e-11	84.132672	-29.408563
100	[1, 0, 0]	axial	1.836152673426	1.836152673337	-8.872e-11	355.201792	+1.869279
101	[1, 0, 1]	face diagonal	1.974587075075	1.974587074999	-7.638e-11	326.873755	-8.694352
111	[1, 1, 1]	body diagonal	2.000023101888	2.000023101807	-8.127e-11	3.082265	-28.568338
01-1	[0, 1, -1]	face diagonal	2.119131490765	2.119131490749	-1.617e-11	81.407976	-8.126533
10-1	[1, 0, -1]	face diagonal	2.200890538669	2.200890538590	-7.884e-11	20.583535	+10.955543
1-1-1	[1, -1, -1]	body diagonal	2.223234046131	2.223234046047	-8.374e-11	348.090587	+28.951893
1-10	[1, -1, 0]	face diagonal	2.272813025845	2.272813025763	-8.176e-11	322.437787	+18.411206
110	[1, 1, 0]	face diagonal	2.273307476082	2.273307476001	-8.143e-11	27.349970	-15.252461
1-11	[1, -1, 1]	body diagonal	2.718195812855	2.718195812787	-6.769e-11	304.223020	+7.599043

Code	Integer h	Class	Reported s_h	Computed s_h	Difference	Reciprocal RA	Reciprocal Dec
11-1	[1, 1, -1]	body diagonal	2.887171860755	2.887171860683	-7.203e-11	41.881515	-4.761634

8.1 Radial law

$$s_h = \sqrt{(h^T G^{-1} h) / (e_3^T G^{-1} e_3)}.$$

The radial unit is a metric norm, not an arbitrary scalar label. The sign pattern of h determines which off-diagonal reciprocal terms add and which subtract. The full list is therefore sensitive to the complete reciprocal geometry, including its handed angular relationships.

8.2 Projective classification

Axial codes contain one nonzero coordinate, face-diagonal codes contain two, and body-diagonal codes contain three. Antipodal identification removes the sign-reversed duplicate but preserves mixed-sign states such as 10-1 and 1-11. Those mixed states are required to recover the off-diagonal sign structure of the reciprocal metric.

9. The Mass Bridge is the axial sub-spectrum of the reciprocal shell

Reciprocal basis norms are direct face areas divided by Cell3 volume:

$$\begin{aligned} ||a^*|| &= A_{bc} / V \\ ||b^*|| &= A_{ac} / V \\ ||c^*|| &= A_{ab} / V. \end{aligned}$$

Normalizing the three axial reciprocal states by code 001 gives

$$\begin{aligned} s_{100} &= A_{bc}/A_{ab} = 1.836152673337283 \\ s_{010} &= A_{ac}/A_{ab} = 1.339905341976038 \\ s_{001} &= 1 \\ s_{100}/s_{010} &= A_{bc}/A_{ac} = 1.370359991721057 \end{aligned}$$

Mass Bridge as the axial reciprocal-shell sub-spectrum

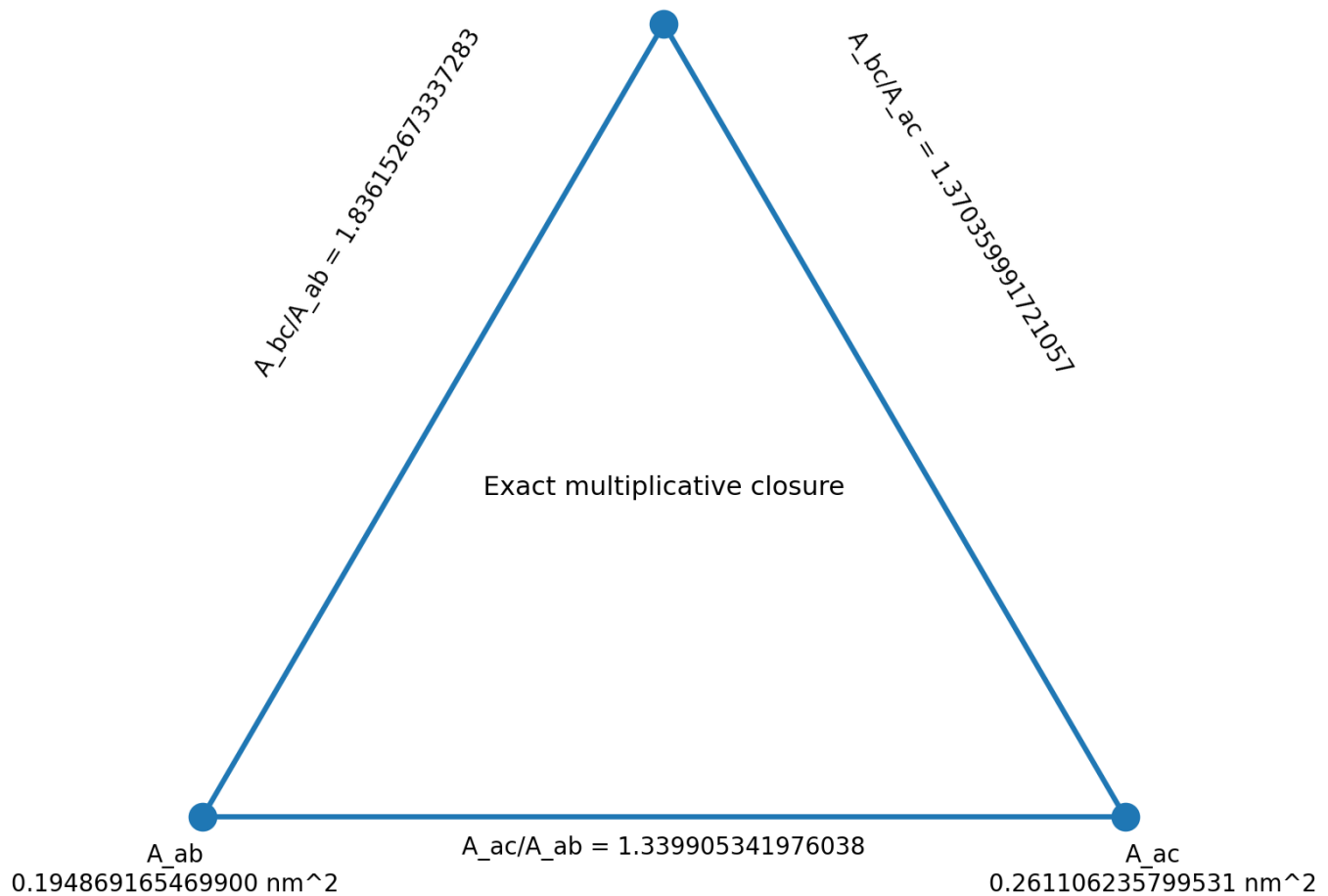


Figure 8. Authoritative Cell3 face-channel triangle. The three displayed ratios are exactly the three axial reciprocal-shell relations, and their multiplicative closure is machine zero.

The Mass Bridge, reciprocal plane spacing, and MOD radial grammar are therefore one mathematical object viewed through different derivative channels. Matter-related ratios use the scale-free face-area form; projective plane analysis uses the reciprocal-normal form; MOD analysis uses the normalized radial form.

Unified identity. Mass Bridge subset reciprocal ternary shell equals axial reciprocal-metric spectroscopy.

10. Reciprocal-metric reconstruction from the shell

A symmetric 3×3 metric has six independent components up to a common normalization. Six radial shell values are sufficient to reconstruct those components. The remaining seven states become redundant closure tests.

$$\begin{aligned}
 R_{11}/R_{33} &= s_{100}^2 \\
 R_{22}/R_{33} &= s_{010}^2 \\
 R_{33}/R_{33} &= 1 \\
 R_{12}/R_{33} &= (s_{110}^2 - s_{100}^2 - s_{010}^2)/2 \\
 R_{13}/R_{33} &= (s_{101}^2 - s_{100}^2 - 1)/2 \\
 R_{23}/R_{33} &= (s_{011}^2 - s_{010}^2 - 1)/2
 \end{aligned}$$

The 13 radial units reconstruct the complete reciprocal metric

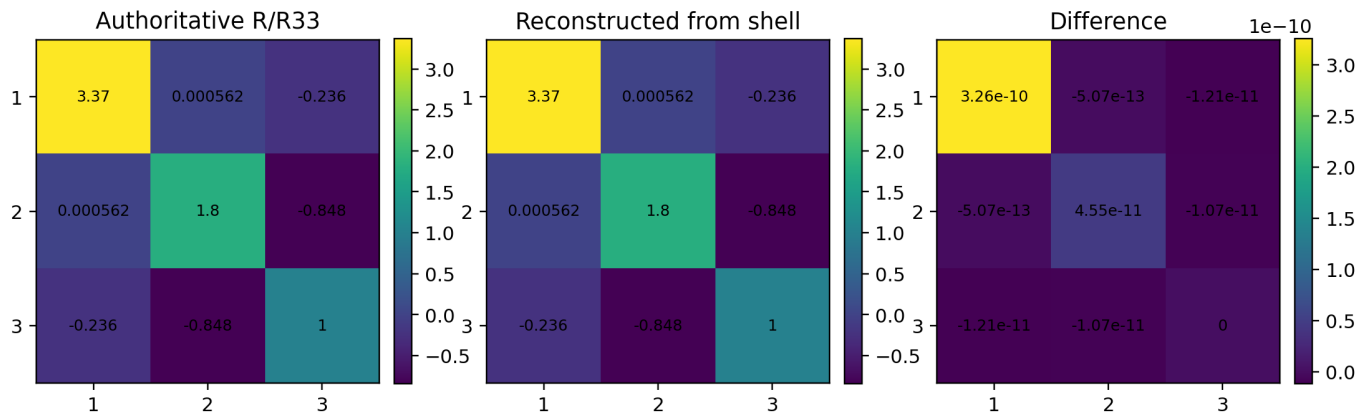


Figure 9. Authoritative normalized reciprocal metric, reconstruction from the radial shell, and component difference. Maximum absolute component error is 3.258e-10, set by decimal truncation of the radial ledger.

The other seven codes verify the signed combinations of the reconstructed entries. A valid MOD radial assignment must therefore satisfy a complete set of quadratic consistency relations. The shell is a redundant metric fingerprint, not a freely adjustable collection of thirteen numbers.

Interpretation. The thirteen-state grammar is reciprocal-metric spectroscopy. It identifies a complete metric state through a finite projective shell.

11. Rank-two binary-ternary scale algebra

The foundational hierarchy contains two independent scale generators. Collective grouping multiplies the basis by 3, while internal subdivision divides it by 2. The complete similarity orbit is

$$B(p, n) = 3^p 2^{-n} B$$

$$\ln(\text{scale}) = p \ln(3) - n \ln(2).$$

A one-dimensional $3/2$ ladder is one diagonal through this two-dimensional integer lattice. Any binary-ternary ratio $r = 2^a 3^b$ can be rewritten as

$$r = 2^{\sigma} (2/3)^{\nu}$$

$$\nu = -b$$

$$\sigma = a + b.$$

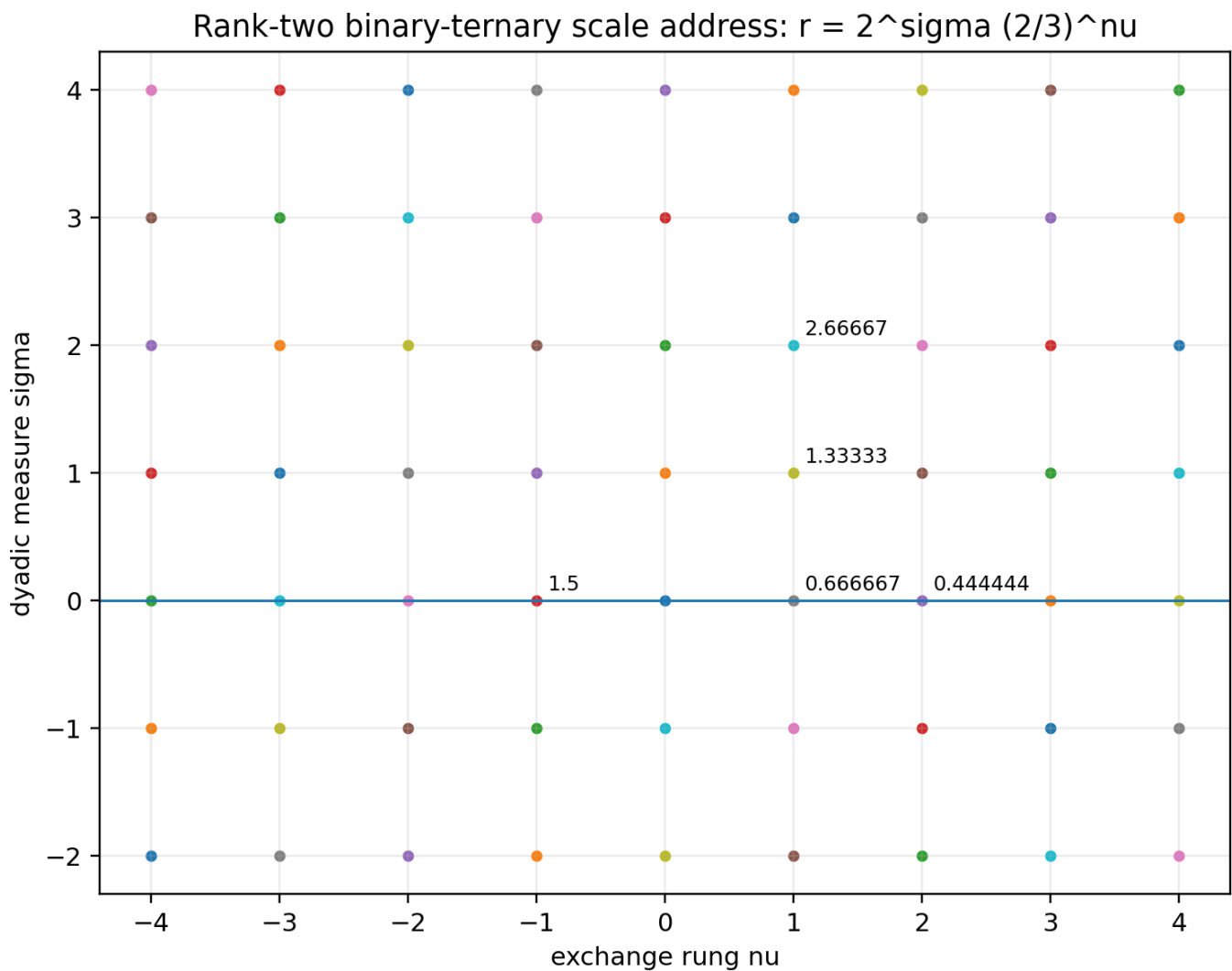


Figure 10. Rank-two scale-address lattice. The line $\sigma = 0$ is the pure $2/3$ exchange ladder; nonzero σ records an independent dyadic measure displacement.

Ratio	ν	σ	Interpretation
$2/3$	1	0	One downward exchange rung
$4/9$	2	0	Two downward exchange rungs
$8/27$	3	0	Three downward exchange rungs
$3/2$	-1	0	One upward exchange rung
$4/3$	1	1	Same exchange rung with one dyadic measure shift
$8/3$	1	2	Same exchange rung with two dyadic measure shifts

11.1 Consequence for solar-system ratios

The Saturn-Neptune-Earth free-rotation sequence lies on the $\sigma = 0$ temporal diagonal. The repeated $4/3$ and $8/3$ mass relations occupy the same exchange rung $\nu = 1$ but different σ states. Bodies that do not lie on one universal $3/2$ line may therefore occupy displaced measure families rather than being outside the recursive system.

11.2 Dimensional measure

One geometric length transformation by $2/3$ induces area and volume transformations by $(2/3)^2$ and $(2/3)^3$. The coordinate ν can therefore also encode tensor or measure degree when the physical observable is a length, area, volume, action, or higher-order carrier. The σ coordinate separates additional dyadic partitioning from the ternary-binary exchange rung.

12. Seven sheets and the hidden A2 tangential coordinates

The 27 Lucas half-step nodes have integer addresses (i,j,k) with each component in $\{-1,0,+1\}$. Define the sheet coordinate and two relative coordinates

$$\begin{aligned}\lambda &= i + j + k \\ x_{i1} &= i - j \\ x_{i2} &= j - k.\end{aligned}$$

The inverse map is

$$\begin{aligned}i &= (\lambda + 2 x_{i1} + x_{i2})/3 \\ j &= (\lambda - x_{i1} + x_{i2})/3 \\ k &= (\lambda - x_{i1} - 2 x_{i2})/3.\end{aligned}$$

Seven-sheet decomposition and the two tangential A2 coordinates

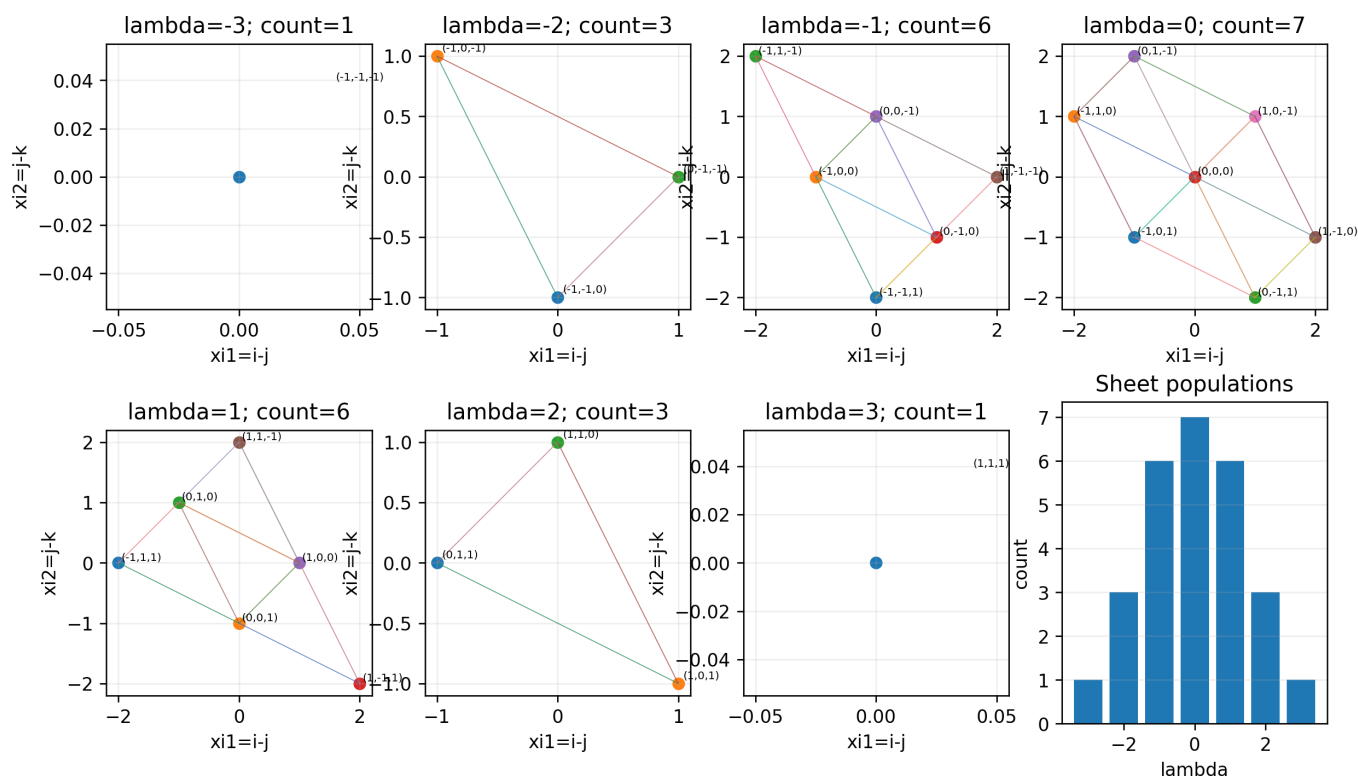


Figure 11. Exact seven-sheet decomposition. Every node is labelled by its native address. In-sheet adjacency is generated by integer displacements whose components sum to zero, producing the A2 tangential kernel.

The populations 1, 3, 6, 7, 6, 3, 1 are the coefficients of $(1 + x + x^2)^3$. Adjacent sheets alternate parity $\chi = (-1)^\lambda$. A state therefore has one laminar coordinate λ and two tangential shear or slip coordinates x_{i1} and x_{i2} .

Unit-sphere projection can preserve the final direction while discarding the separate amplitudes of these three internal coordinates. Angular bands can consequently contain multiple distinct native addresses that are only partially resolved in phase space.

13. Exterior grade as a hidden discrete dimension

Three-dimensional projection identifies several geometrically different objects with ordinary direction vectors. Hodge duality maps a plane bivector into a pseudovector, so plane normals and direct vectors can be plotted on the same sphere even though they belong to different exterior grades.

Grade g	Geometric object	Representative physical rendering
0	Scalar occupation	Mass density, localized occupation, scalar carrier

Grade g	Geometric object	Representative physical rendering
1	Direct vector	Displacement, transport, velocity, force direction
2	Bivector / reciprocal plane	Plane normal, circulation, torque plane, gradient response
3	Oriented volume	Chirality, handed Cell3 volume, pseudoscalar orientation

The exterior grade g in $\{0,1,2,3\}$ is therefore a required state label. Two observations can share identical RA and Dec while representing different grades. Mixing body axes, plane normals, torque axes, gradient directions, and crossing rays without this label creates apparent extra branches that are artifacts of projection rather than distinct foundations.

The exact exterior closure in Chapter 6 is the transition between grades 2 and 1: two reciprocal planes intersect along a direct ray. The converse transition maps two direct rays into a reciprocal plane.

14. Phase registry as an attached fiber

Spatial address does not exhaust the state. The quantized hierarchy carries an additional residual coordinate obtained after coarse geometric and carrier content has been removed. In the recursive coordinate $\rho = 3/2$, a residual ratio q has

$$x_{\text{epsilon}} = \log_{(3/2)}(|q|) = \text{integer depth} + \text{fractional phase}.$$

The signed state retains polarity, integer depth, and an admissible phase well. A fractional phase can recur at different depths without becoming the same complete state. The current solar-system MOD reduction uses $N = 81 = 3^4$, naturally resolving four ternary registry levels.

The phase coordinate is not another Cartesian spatial axis. It is a fiber attached to each spatial, reciprocal, or projective address. The complete state space therefore has the form

$$\text{spatial/projective address} \times \text{phase/registry address}.$$

A dominant $q = 0$ class means that many observations occupy the central residual phase class after their metric, shell, numerator, denominator, scale, and branch addresses have been assigned. It does not mean those observations are physically identical.

15. Solar-system rotation, mass, and orbital branches

15.1 Rotation branches

The 31-body rotation ledger separates into dynamical branches. Twenty regular and large satellites are synchronous, Mercury occupies a 3:2 spin-orbit branch, and the remaining free rotators occupy intrinsic rotational states. Tidal and resonance dynamics select the branch; the K-Field supplies the deeper address grammar within and between branches.

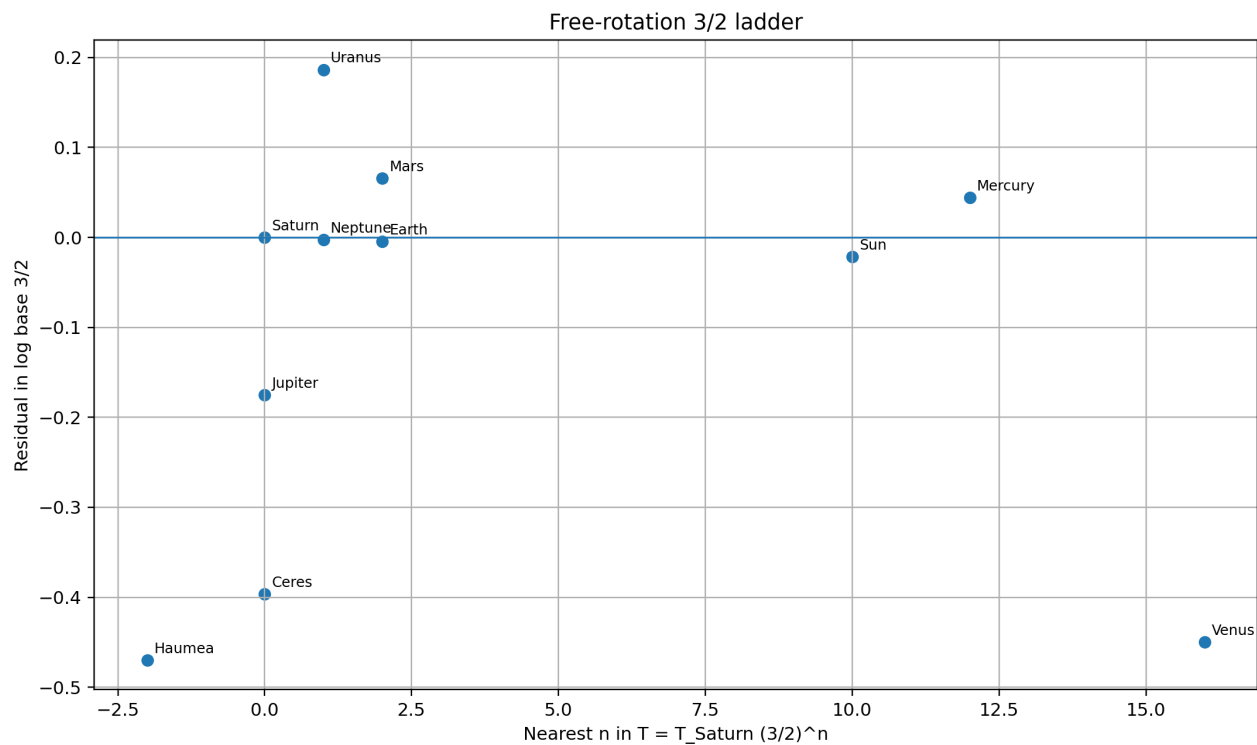


Figure 12. Actual Saturn-referenced free-rotation ladder. Saturn, Neptune, and Earth occupy consecutive 2/3 period rungs; Mars, the Sun, and Mercury approach higher or displaced states.

Relation	Observed ratio	Target	Relative residual (%)
Europa:Io	2.007294785	2.000000000	+0.364739
Ganymede:Europa	2.014696790	2.000000000	+0.734839
Ganymede:Io	4.044090359	4.000000000	+1.102259
Tethys:Mimas	2.003139720	2.000000000	+0.156986
Dione:Enceladus	1.997430617	2.000000000	-0.128469
Pluto:Charon	1.000000000	1.000000000	+0.000000
Neptune:Saturn	1.498307718	1.500000000	-0.112819
Earth:Neptune	1.499061848	1.500000000	-0.062543
Earth:Saturn	2.246055936	2.250000000	-0.175292

Synchronous locking copies the orbital temporal address: $n_{\text{rot}} = n_{\text{P}}$. Mercury shifts that orbital address by one downward exchange rung because $T_{\text{rot}} = (2/3) P$. The Saturn-Neptune-Earth ladder applies the same -1 shift successively to free-rotation states.

15.2 Mass partitions

The mass spectrum does not collapse to one monotonic ladder. It contains local binary-ternary partition families, including repeated 4/3 and 8/3 relations. In the rank-two scale algebra, these share exchange rung $\nu = 1$ while differing in σ . The recurrence across different parent systems indicates a common scale grammar rather than a parent-specific numerical accident.

Mass relation	Observed ratio	Binary-ternary target	Residual (%)
Ariel / Ceres	1.333169434	4/3	-0.012292
Titania / Umbriel	2.666274971	8/3	-0.014689
Oberon / Rhea	1.333621321	4/3	+0.021599

15.3 Known orbital mechanics as address transport

The primitive orbital quantities are mass m , parent gravitational parameter μ , semimajor axis a , eccentricity e , and branch selection. Known orbital mechanics transports these primitive states into period, speed, acceleration, angular momentum, and energy:

$$P = 2 \pi a^{(3/2)} \mu^{(-1/2)}$$

$$v = \mu^{(1/2)} a^{(-1/2)}$$

$$a_c = \mu a^{(-2)}$$

$$L = m \mu^{(1/2)} a^{(1/2)} \sqrt{1-e^2}$$

$$K = (1/2) m \mu a^{(-1)}.$$

In a logarithmic 3/2 address, multiplication becomes addition and exponentiation becomes scalar multiplication. These equations are therefore exact linear address maps. They preserve known mechanics while exposing the deeper scale coordinates carried by each observable.

16. Hidden empirical dimensions of the orbital ledger

Seven logarithmic orbital observables were standardized across the 30 orbiting bodies: mass, orbital period, semimajor axis, mean orbital speed, centripetal acceleration, total orbital angular momentum, and mean orbital kinetic energy. Their population covariance eigensystem contains only three substantial dimensions.

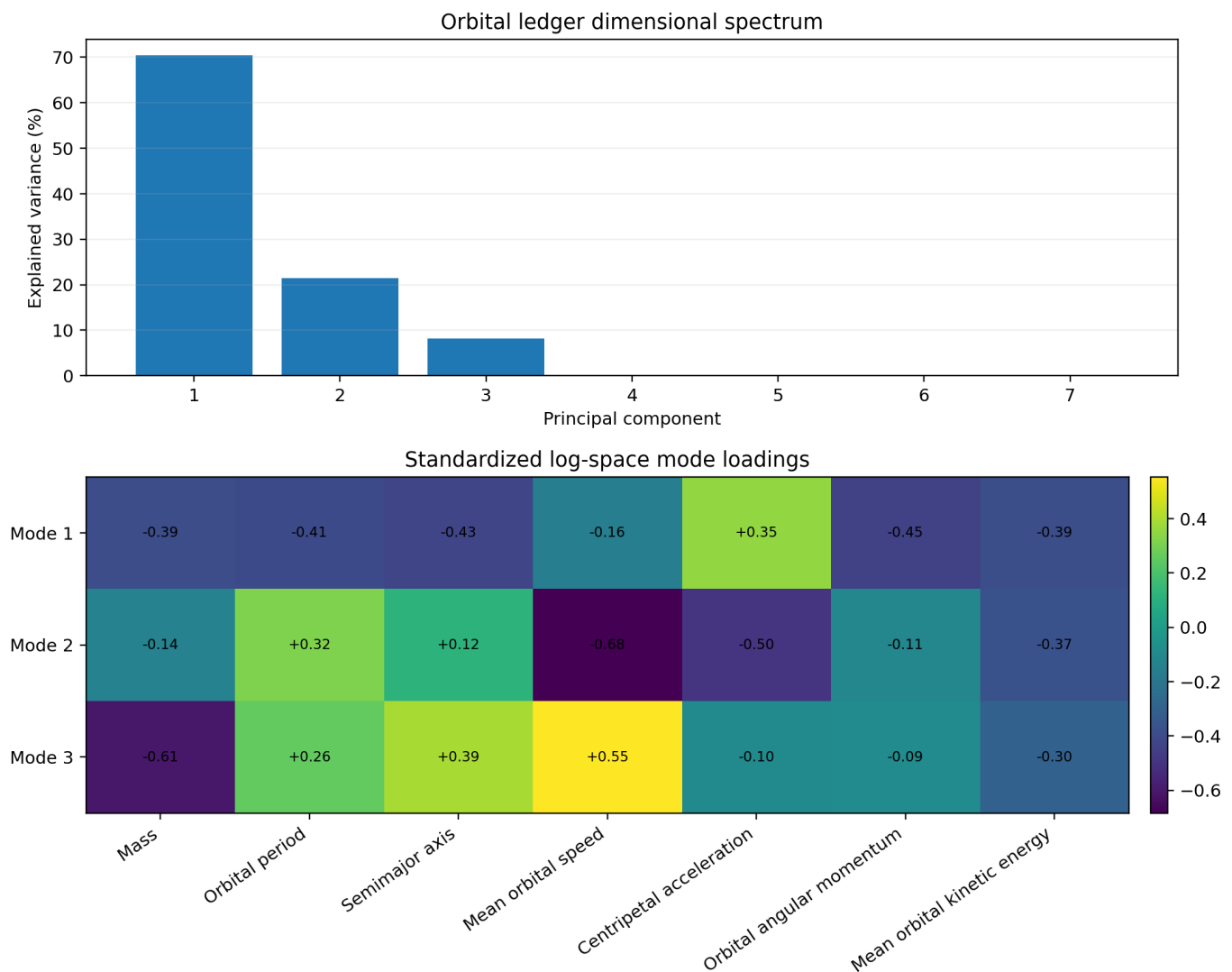


Figure 13. Orbital-ledger principal components computed from standardized natural logarithms. The first three modes account for essentially the complete seven-variable population.

Mode	Eigenvalue	Explained variance (%)	Physical interpretation
1	4.92571482864	70.3673547	System scale/action: mass, extent, period, angular momentum, and energy
2	1.50061734585	21.4373907	Orbital intensity/curvature: velocity and acceleration relative to period and radius
3	0.573667487521	8.19524982	Occupation-velocity partition: redistribution among mass, velocity, and energy
4	3.37985753525e-07	4.82836791e-06	Eccentricity/shape defect entering primarily through angular momentum

The first three modes account for 70.3674%, 21.4374%, and 8.19525% of the variance. The fourth mode has eigenvalue approximately 3.38×10^{-7} and represents the weak eccentricity correction entering L through $\sqrt{1-e^2}$. Remaining modes are numerical zero because the derived quantities satisfy exact algebraic closures.

16.1 Interpretation

The measured seven-dimensional ledger is a rendered image of approximately three dominant physical state dimensions plus one weak shape coordinate. This is an empirical manifestation of explanatory compression: several conventional observables are not independent degrees of freedom, but known transformations of a smaller primitive state.

17. Orbital closure nullspace and the Jack derivative layer

The definitions used in the ledger imply five exact closure identities:

$$\begin{aligned} P v / a &= 2 \pi \\ P^2 a_c / a &= 4 \pi^2 \\ P a_c / v &= 2 \pi \\ K &= (1/2) m v^2 \\ L &= m a v \sqrt{1-e^2}. \end{aligned}$$

Taking logarithms or recursive addresses converts these identities into linear equations. If n_X is the 3/2 address of quantity X , then

$$\begin{aligned} n_P + n_v - n_a &= n(2 \pi) \\ 2 n_P + n_{ac} - n_a &= n(4 \pi^2) \\ n_P + n_{ac} - n_v &= n(2 \pi) \\ n_K - n_m - 2 n_v &= n(1/2) \\ n_L - n_m - n_a - n_v - (1/2)n(1-e^2) &= 0. \end{aligned}$$

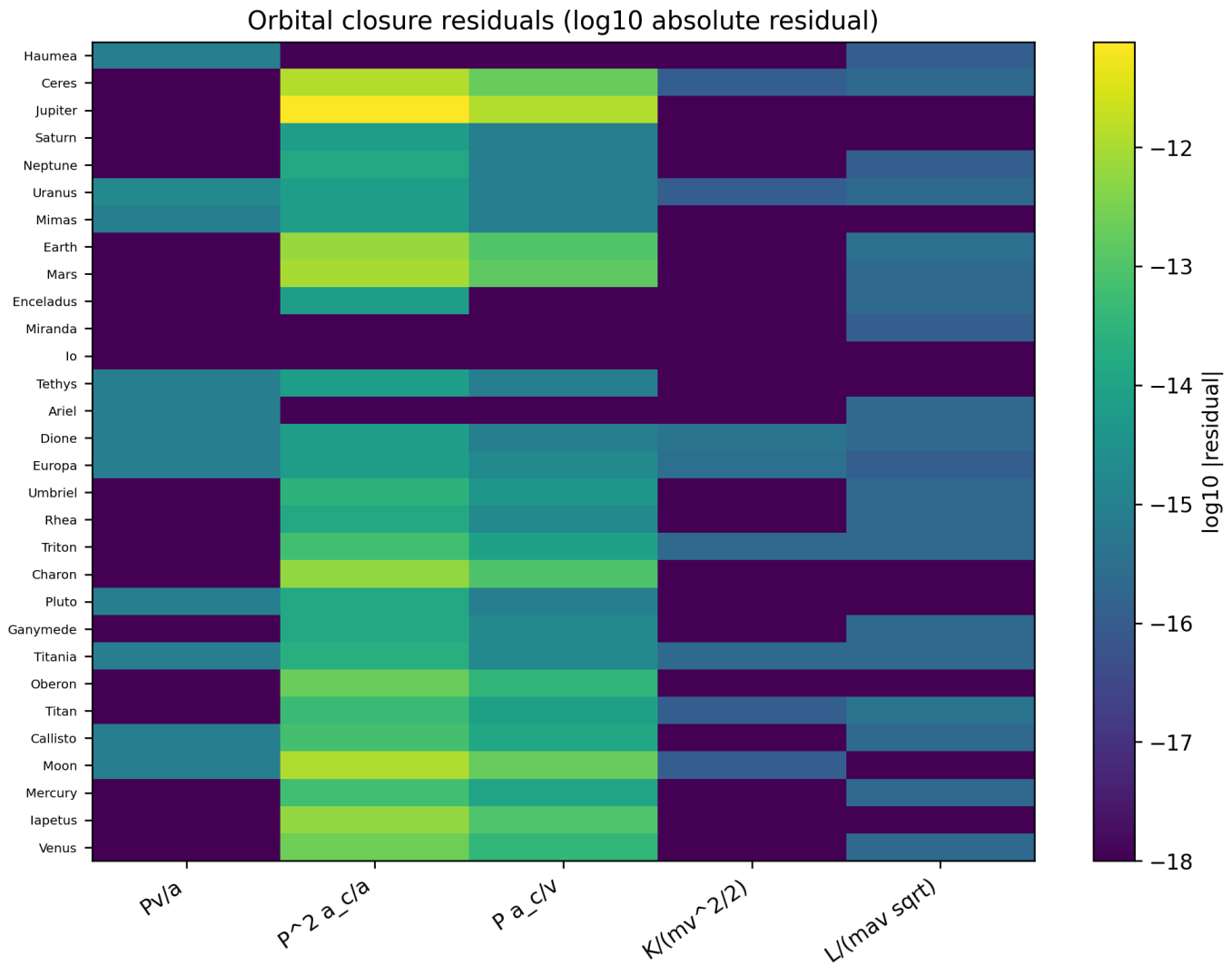


Figure 14. Numerical closure residuals for every orbiting body. Values cluster at floating-point precision because the kinematic quantities were generated from the same exact definitions.

These equations define an affine nullspace $C_n = c$. The Jack layer is the closure Jacobian that maps primitive K-Field variation into externally rendered observables. A physically meaningful MOD continuation must fit the primitive addresses and predict the derived addresses through this closure map. Independently fitting every derived quantity introduces redundant freedom and obscures the residual phase.

Closure criterion. Observed composite address equals the address predicted from primitive states plus a small residual phase.

18. Quadratic phase metric and the six-node intersection skeleton

Directional transition cost on the unit sphere is represented by the symmetric positive-definite matrix G_{phase} . For a unit direction n ,

$$q(n) = n^T G_{\text{phase}} n$$

$$f_G(n) = \sqrt{q(n)}.$$

Its orthonormal eigenaxes are the global soft, intermediate, and hard directions. A maximum-modulation great circle contains the soft and hard axes and has the intermediate axis as its plane normal. Two distinct zero-modulation great circles contain the intermediate axis and remain at constant cost $\sqrt{\lambda_2}$.

Phase metric: three eigenaxes, three extremal planes, six intersection nodes

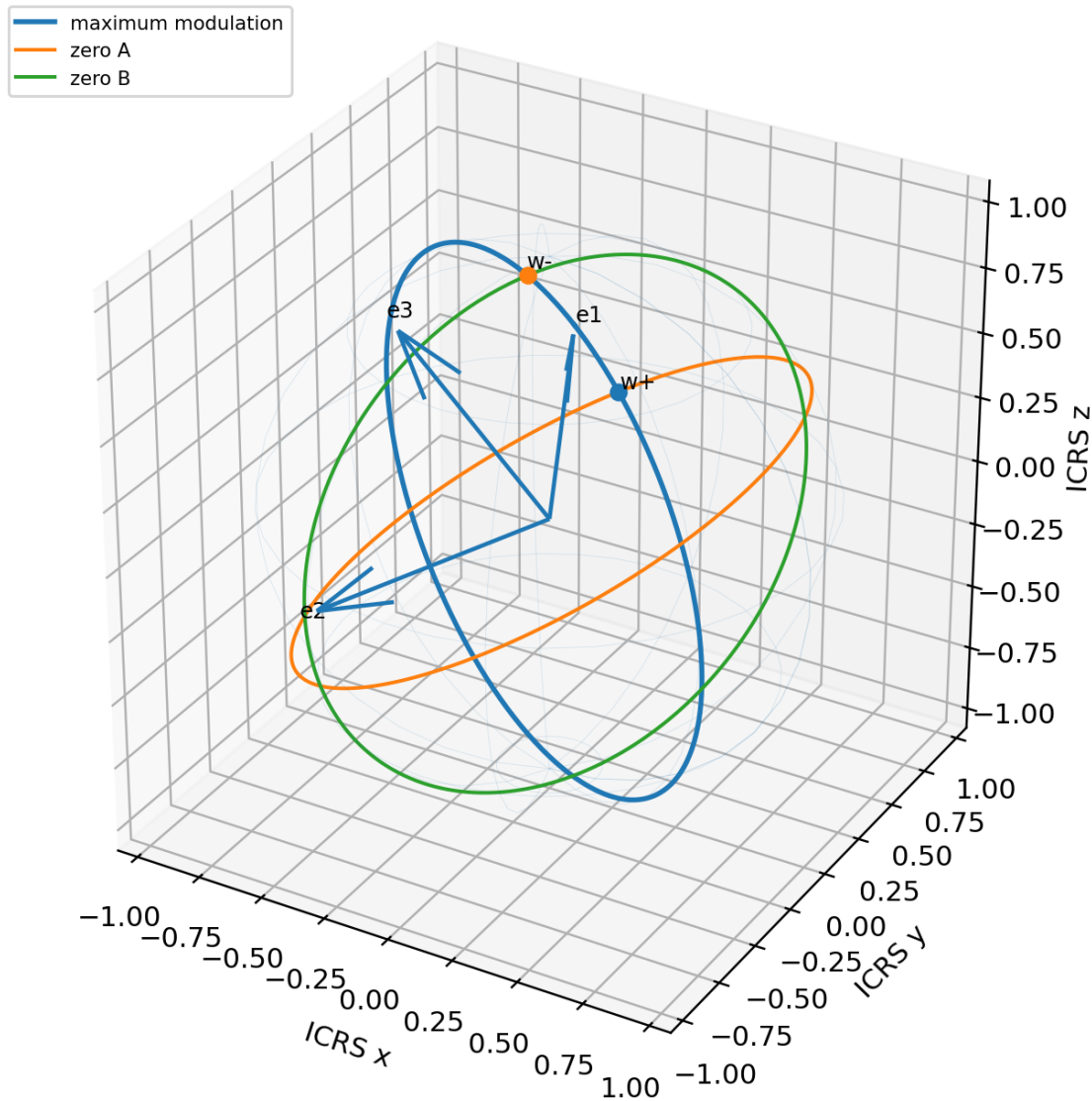


Figure 15. Actual phase metric on the unit sphere. The maximum-modulation circle joins the soft and hard axes. The two zero-modulation circles intersect it at four additional nodes and intersect each other at the intermediate axis pair.

Mode	Eigenvalue	Cost sqrt(lambda)	RA (deg)	Dec (deg)	Role
1	1.429328753487483	1.195545379100051	114.813083651	+12.969895704	Global minimum / soft axis
2	2.112545026447479	1.453459674861150	197.222212119	-29.836687522	Intermediate axis
3	8.800336434665038	2.966536100347514	225.519007199	+56.919969623	Global maximum / hard axis

18.1 Exact zero-modulation construction

$a = \sqrt{(\lambda_2 - \lambda_1) / (\lambda_3 - \lambda_1)} = 0.304449783157334$
 $b = \sqrt{(\lambda_3 - \lambda_2) / (\lambda_3 - \lambda_1)} = 0.952528387784560$
 $m_{\pm} = a e_1 \pm b e_3$
 $w_{\pm} = b e_1 \mp a e_3$

The two zero-modulation normals have projective dihedral angle 35.450129229 deg. Their planes intersect at $\pm e_2$. Each zero plane intersects the maximum-modulation plane at one additional antipodal pair, $\pm w_+$ or $\pm w_-$. The complete extremal-plane arrangement therefore contains six exact intersection nodes.

Node pair	RA (deg)	Dec (deg)	Geometric role
e2	197.222212119	-29.836687522	Projective pair including antipode

Node pair	RA (deg)	Dec (deg)	Geometric role
w_plus	105.863163197	-2.367942424	Projective pair including antipode
w_minus	124.948934124	+27.961995685	Projective pair including antipode

18.2 Candidate direct-shell soft-axis lock

Candidate soft-axis lock: angular separation 1.852756 deg

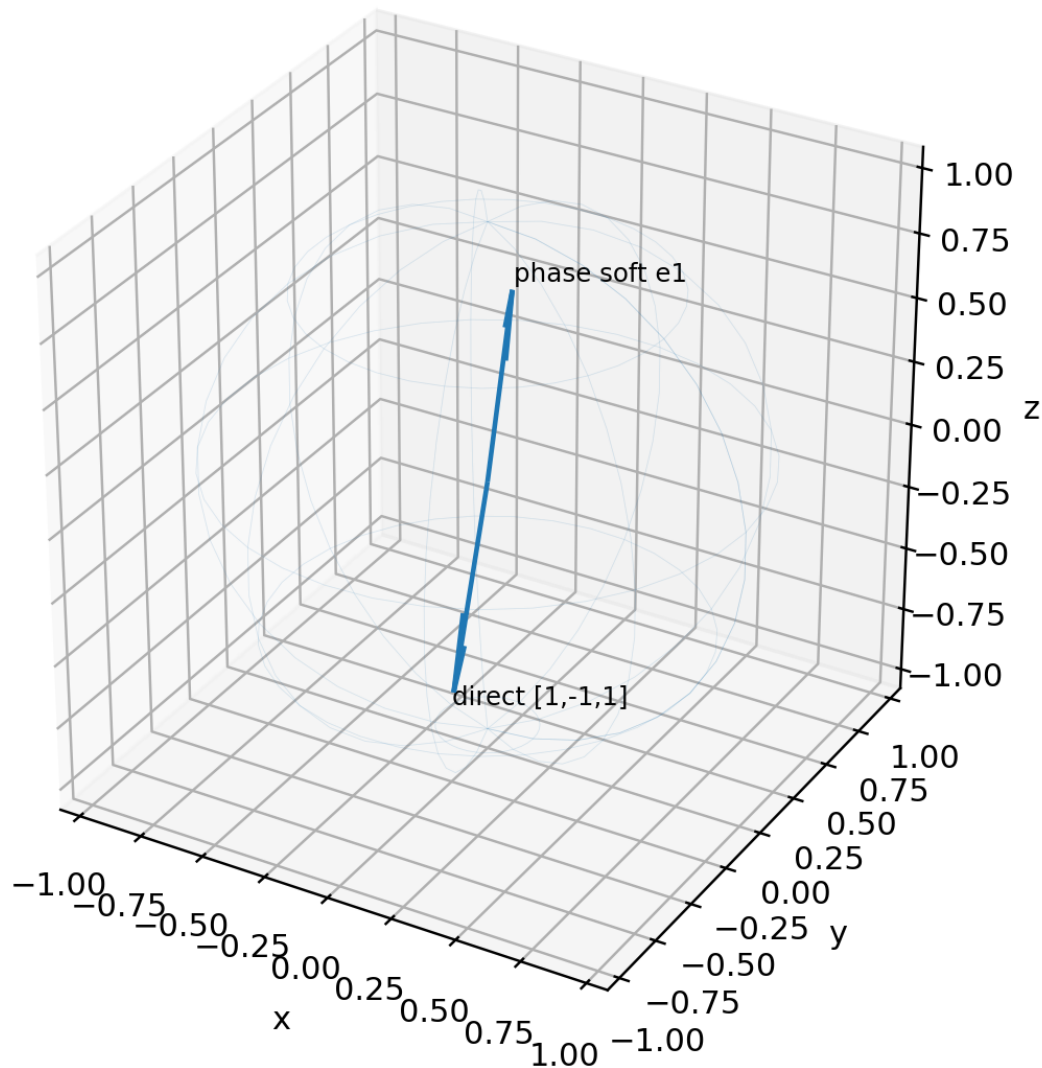


Figure 16. The phase soft eigenaxis and the direct first-shell address $[1,-1,1]$ are separated by 1.852756 deg. The intermediate and hard axes do not show comparably close first-shell direct matches.

The soft-axis relation is a candidate lock between the phase metric and the direct first shell. It identifies a specific low-complexity direct ray for deeper analysis. A complete shell identification requires the higher-address search for the intermediate and hard axes and the same direct/reciprocal grade separation used elsewhere in this work.

19. Structural eigensystem of the assimilated corpus

The corpus graph contains 45 conceptual nodes and 635 directed relations assembled from 26 source works. Cell3 and K-Field are the dominant hubs. The most informative nodes are the bridges connecting otherwise different domains: global orientation, polarity, fine-structure branch, Mass Bridge, interaction hierarchy, scale bridge, reciprocal geometry, and structure-interaction duality.

Assimilated corpus graph: dominant structural bridge nodes

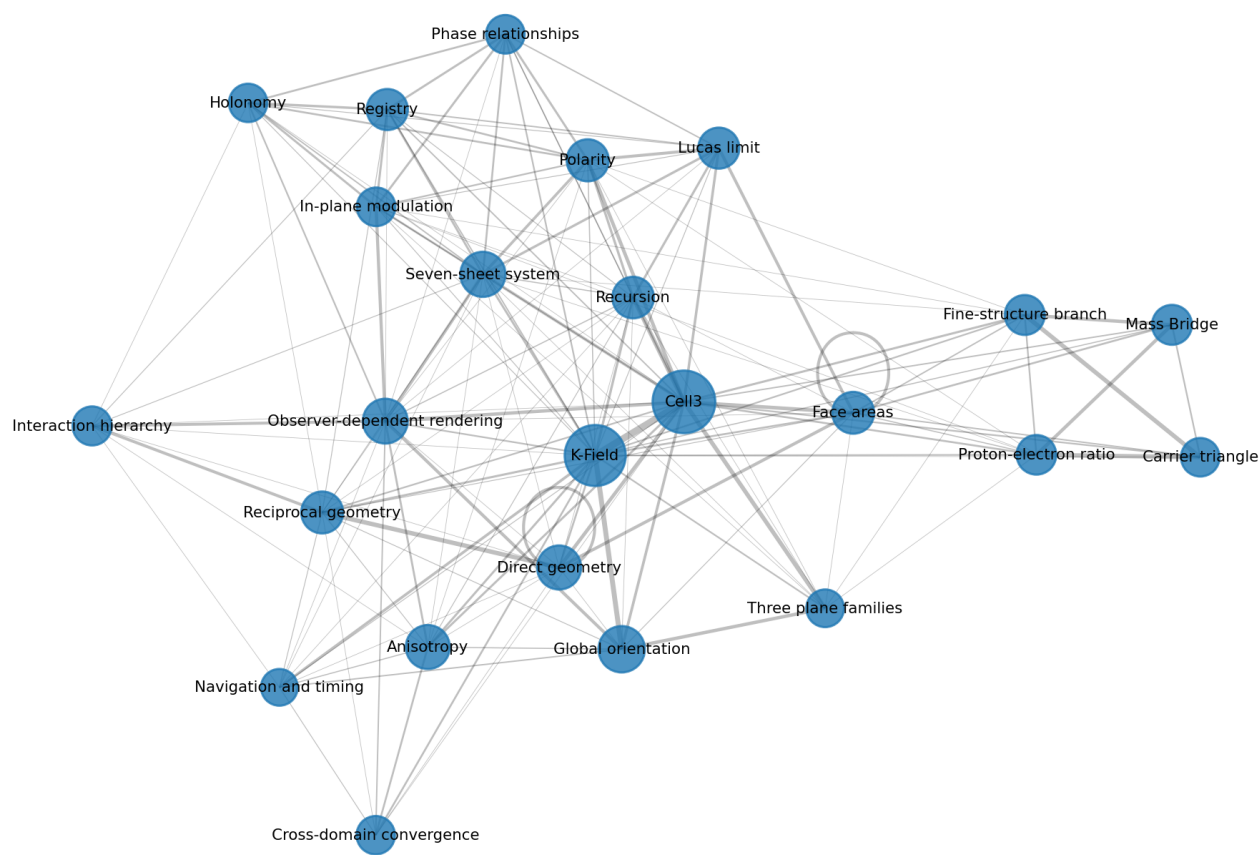


Figure 17. Top corpus nodes by PageRank with actual weighted co-occurrence and dependency edges. Node size is proportional to PageRank; the layout is deterministic and used only to expose bridge structure.

The normalized corpus adjacency has four stable nontrivial structural modes. Their persistence under multiplicative edge reweighting identifies the conceptual dimensions repeatedly required to connect the independent work.

Mode	Eigenvalue	Structural content	Median alignment
1	0.73124	Mass Bridge, carrier triangle, proton/electron relation, fine structure, reduced-mass chain	0.99859
2	0.64696	Registry, holonomy, phase, polarity, stiffness, seven-sheet memory	0.99839
3	0.54861	Direct-reciprocal duality, interaction hierarchy, structure-interaction mapping	0.99603
4	0.49480	Fixed global orientation versus sidereal, lunar, and oscillator modulation	0.99516

These corpus modes correspond closely to four physical derivative classes: scalar carrier, phase memory, direct/reciprocal interaction, and observer modulation. They are conceptual eigendimensions of the integrated architecture, not replacements for the physical phase-metric eigenvalues.

20. Minimal extended K-Field state

The complete architecture can be represented by the state tuple

$S = (g, [z], p, n, \lambda, \xi_1, \xi_2, \chi, \phi, \mu).$

Coordinate	Meaning	What projection can hide
g	Exterior grade 0,1,2,3	Distinguishes scalar, vector, plane/bivector, and oriented volume
$[z]$	Projective integer direction	Integer numerator scale and antipodal orientation
p,n	Rank-two scale address	Absolute recursive rung and separate binary/ternary generator counts

Coordinate	Meaning	What projection can hide
lambda	Laminar sheet coordinate	Normal position in the seven-sheet structure
xi1,xi2	Tangential A2 coordinates	In-sheet shear, slip, and relative occupation
chi	Polarity or chirality	Mirror branch and alternating sheet parity
phi	Phase/registry coordinate	Residual well and recursive phase memory
mu	Occupation or amplitude	Population, mass carrier, and multiplicity

The observer sees only a partial rendering. A direct angular observation is $\pm B_z/|B_z|$; a reciprocal angular observation is $\pm B^{-T} z/|B^{-T} z|$. Scale, grade, sheet, phase, and occupation are suppressed. They reappear through period ratios, masses, orbital actions, resonance branches, intersection multiplicity, polarity, and directional modulation.

Minimum architecture. Projective integer direction, exterior grade, two scale generators, three sheet/tangential coordinates, polarity, phase registry, and occupation are sufficient to organize the recurring angular, ratiometric, orbital, material, and plane-intersection results.

21. Derived predictions and fitting program

21.1 Galaxy crossing back-projection

For every high-multiplicity projective crossing c , compute $B^{-1} c$ and identify the nearest low-complexity integer ray. The observed plane multiplicity should increase with arithmetic representation degeneracy D_H after controlling for aperture and plane count.

21.2 Direct-versus-reciprocal angular families

Spin axes, displacement axes, and occupancy directions are to be compared with B_m . Plane normals, gradients, and reciprocal responses are to be compared with $B^{-T} h$. The exterior-grade label must be assigned before combining catalogs.

21.3 Radial-shell closure

Every valid thirteen-state assignment must reconstruct one normalized reciprocal metric. Seven redundant shell states then test the off-diagonal signs and higher mixed combinations. Failure of shell closure identifies a code, sign, normalization, or source mismatch.

21.4 Rank-two ratio mapping

Rotation, mass, length, area, volume, action, and energy ratios are to be fitted on the (ν, σ) lattice. The pure $3/2$ family is the $\sigma = 0$ diagonal, not the complete scale space.

21.5 Orbital Jack closure

Fit mass, parent gravitational parameter, radius, eccentricity, and branch address. Predict period, speed, acceleration, angular momentum, and energy through the exact closure equations. Compare only the remaining residual phase with K-Field orientation, phase cost, curvature, and Cell3-plane distance.

21.6 Distance-dependent plane convergence

Parallel physical structures assigned to one reciprocal family should display angular displacement β proportional to $1/r$. The fitted physical plane offset t should remain constant when transformed back from angular observations at different distances.

21.7 Six-node phase skeleton

Dynamical axes and plane crossings should be compared with $\pm e^2$, $\pm w^+$, and $\pm w^-$. The six-node skeleton is the complete intersection structure of the three extremal phase planes and provides a sharper target than the three eigenaxes alone.

22. Worked derivations

22.1 Deriving a direct projective angle

Given two integer addresses m and n , form the three scalars $m^T G m$, $n^T G n$, and $m^T G n$. Their normalized quotient is the cosine of the physical angular separation. No floating three-dimensional reconstruction is required beyond the fixed metric.

Example procedure:

1. choose integer rays m and n ;
2. evaluate $Q_m = m^T G m$ and $Q_n = n^T G n$;
3. evaluate $C_{mn} = m^T G n$;
4. $\theta = \arccos(C_{mn} / \sqrt{Q_m Q_n})$.

22.2 Recovering a reciprocal metric entry

The axial shell values give R_{11}/R_{33} and R_{22}/R_{33} . The 110 diagonal adds R_{12} twice. Solving for the cross term gives

$$R_{12}/R_{33} = [s_{110}^2 - s_{100}^2 - s_{010}^2]/2.$$

The same procedure with 101 and 011 recovers R_{13} and R_{23} . Signed diagonal states then verify the result.

22.3 Mapping a mass ratio onto the scale lattice

For $r = 4/3$, write $r = 2^{1/2} 3^{-1}$. Then $\nu = 1$ and $\sigma = 1$. For $r = 8/3$, write $r = 2^{2/3} 3^{-1}$. Then $\nu = 1$ and $\sigma = 2$. The shared ν shows one common exchange rung; σ distinguishes the dyadic measure state.

22.4 Back-projecting a plane crossing

Given two reciprocal integer addresses h and k , compute $m = h \times k$. The physical crossing ray is B_m . If the plane normals are measured rather than integer-labelled, first fit h and k in the reciprocal metric, then apply the exact integer cross product.

22.5 Closing an orbital state

Given m , a , e , and parent μ , calculate P , v , a_c , L , and K from the standard equations. Convert each to its recursive address. The linear closure residuals must vanish apart from source precision and any declared K-Field residual phase.

23. Foundational synthesis

The K-Field is a recursive, oriented, anisotropic projective exterior lattice. Cell3 supplies the invariant local shape. The Lucas Group supplies the collective ternary root. Dyadic subdivision supplies unlimited internal refinement. The direct metric renders displacement and occupancy; the reciprocal metric renders planes, gradients, and directional resolution; exterior algebra closes planes back into rays; the seven sheets supply one laminar and two tangential coordinates; phase registry supplies recursive memory; polarity supplies mirror state; occupation supplies amplitude and multiplicity; and known physical equations provide the closure maps into conventional observables.

The hidden relationships developed in this work are mutually reinforcing:

1. Reciprocal-plane crossings are direct integer rays.
2. The thirteen MOD units are the complete first ternary projective shell.
3. The Mass Bridge is the axial part of that shell.
4. The shell reconstructs the full reciprocal metric.
5. Binary-ternary observables require two scale coordinates.
6. Angular observations require an exterior-grade label.
7. Orbital observables occupy a low-dimensional closure nullspace.
8. Plane-crossing multiplicity is an arithmetic representation degeneracy.
9. The phase metric contributes an exact six-node intersection skeleton.
10. The corpus itself resolves into stable carrier, memory, interaction, and observer modes.

The resulting explanatory chain is

```
integer occupation
-> direct/reciprocal metric
-> exterior-grade transformation
-> projective angular spectrum
-> rank-two scale and phase address
-> known physical closure
-> measured observable.
```

This architecture provides a deeper foundation beneath known science without replacing its laws. The laws act as exact derivative maps of a more complete state. Broad observations across astrophysics, orbital dynamics, matter ratios, angular organization, and force response can therefore be tested as coherent K-Field address fits within one internally closed geometry.

Final determination. The projective exterior lattice with rank-two scale algebra and phase registry is the lowest-dimensional structure presently sufficient to reproduce the recurring geometric and physical relationships across the complete assimilated corpus.

Appendix A. Complete 27-node Lucas ledger

Address	lambda	xi1	xi2	Polarity	x (nm)	y (nm)	z (nm)	Norm (nm)
(-1,-1,-1)	-3	0	0	-1.0	-0.176351350144	0.118682429904	0.637311498326	0.671826810840
(-1,-1,0)	-2	0	-1	1.0	-0.181416008957	-0.076211421278	0.292444294144	0.352482076447
(-1,0,-1)	-2	-1	1	1.0	-0.158685962221	0.222342355125	0.363610955630	0.454785976634
(0,-1,-1)	-2	1	0	1.0	-0.012600729109	0.091233925961	0.618567746877	0.625386652478
(-1,-1,1)	-1	0	-2	-1.0	-0.186480667770	-0.271105272459	-0.052422910037	0.333198543971
(-1,0,0)	-1	-1	0	-1.0	-0.163750621035	0.027448503943	0.018743751448	0.167089839537
(-1,1,-1)	-1	-2	2	-1.0	-0.141020574299	0.326002280346	0.089910412934	0.366398924016
(0,-1,0)	-1	1	-1	-1.0	-0.017665387922	-0.103659925221	0.273700542696	0.293205445207
(0,0,-1)	-1	0	1	-1.0	0.005064658813	0.194893851182	0.344867204181	0.396159882519
(1,-1,-1)	-1	2	0	-1.0	0.151149891926	0.063785422018	0.599823995429	0.621855043707
(-1,0,1)	0	-1	-1	1	-0.168815279848	-0.167445347239	-0.326123452733	0.403600110809
(-1,1,0)	0	-2	1	1	-0.146085233113	0.131108429164	-0.254956791248	0.321765879072
(0,-1,1)	0	1	-2	1	-0.022730046735	-0.298553776403	-0.071166661485	0.307759169052
(0,0,0)	0	0	0	1	0.000000000000	0.000000000000	0.000000000000	0.000000000000
(0,1,-1)	0	-1	2	1	0.022730046735	0.298553776403	0.071166661485	0.307759169052
(1,-1,0)	0	2	-1	1	0.146085233113	-0.131108429164	0.254956791248	0.321765879072
(1,0,-1)	0	1	1	1	0.168815279848	0.167445347239	0.326123452733	0.403600110809
(-1,1,1)	1	-2	0	-1	-0.151149891926	-0.063785422018	-0.599823995429	0.621855043707
(0,0,1)	1	0	-1	-1	-0.005064658813	-0.194893851182	-0.344867204181	0.396159882519
(0,1,0)	1	-1	1	-1	0.017665387922	0.103659925221	-0.273700542696	0.293205445207
(1,-1,1)	1	2	-2	-1	0.141020574299	-0.326002280346	-0.089910412934	0.366398924016
(1,0,0)	1	1	0	-1	0.163750621035	-0.027448503943	-0.018743751448	0.167089839537
(1,1,-1)	1	0	2	-1	0.186480667770	0.271105272459	0.052422910037	0.333198543971
(0,1,1)	2	-1	0	1	0.012600729109	-0.091233925961	-0.618567746877	0.625386652478
(1,0,1)	2	1	-1	1	0.158685962221	-0.222342355125	-0.363610955630	0.454785976634
(1,1,0)	2	0	1	1	0.181416008957	0.076211421278	-0.292444294144	0.352482076447
(1,1,1)	3	0	0	-1	0.176351350144	-0.118682429904	-0.637311498326	0.671826810840

Appendix B. Solar-system body ledger

Body	Parent	Mass (kg)	Orbit a (m)	Orbit P (d)	Mean v (m/s)	Rotation P (d)	Branch
Haumea	Sun	4.00600000e+21	6.45200000e+12	103774	4521.38956	0.1631	free
Ceres	Sun	9.38416000e+20	4.13690250e+11	1681.63	17890.0257	0.378090417	free
Jupiter	Sun	1.89812500e+27	7.78570000e+11	4332.589	13068.2158	0.413538326	free
Saturn	Sun	5.68317000e+26	1.43353000e+12	10759.22	9689.29378	0.444009259	free
Neptune	Sun	1.02409200e+26	4.49506000e+12	60182	5431.69034	0.6652625	free
Uranus	Sun	8.68099000e+25	2.87246000e+12	30688.5	6806.82295	0.718333333	free retrograde
Mimas	Saturn	3.75094017e+19	1.86000000e+08	0.942422	14352.7016	0.94242181	synchronous 1:1
Earth	Sun	5.97217000e+24	1.49598023e+11	365.25636	29784.7661	0.997269632	free
Mars	Sun	6.41691000e+23	2.27939200e+11	686.98	24129.096	1.02595676	free
Enceladus	Saturn	1.08031854e+20	2.38400000e+08	1.370218	12652.6854	1.37021808	synchronous 1:1
Miranda	Uranus	6.44262320e+19	1.29846000e+08	1.413479	6680.4442	1.41347923	synchronous 1:1
Io	Jupiter	8.92964876e+22	4.21800000e+08	1.762732	17401.4891	1.7691378	synchronous 1:1
Tethys	Saturn	6.17495917e+20	2.95000000e+08	1.887802	11364.0124	1.88780256	synchronous 1:1
Ariel	Uranus	1.25106753e+21	1.90929000e+08	2.520379	5508.99238	2.52037887	synchronous 1:1
Dione	Saturn	1.09548672e+21	3.77700000e+08	2.736916	10035.7918	2.73691555	synchronous 1:1
Europa	Jupiter	4.79857378e+22	6.71100000e+08	3.525463	13843.2226	3.55118108	synchronous 1:1
Umbriel	Uranus	1.27504008e+21	2.65986000e+08	4.144177	4667.52452	4.14417625	synchronous 1:1
Rhea	Saturn	2.30648532e+21	5.27200000e+08	4.517503	8486.78261	4.51750262	synchronous 1:1
Triton	Neptune	2.14029256e+22	3.54800000e+08	5.876994	4390.30295	5.87685408	synchronous 1:1 retrograde
Charon	Pluto	1.58967982e+21	1.96000000e+07	6.387222	223.156831	6.387223	mutual synchronous 1:1
Pluto	Sun	1.30245868e+22	5.90638000e+12	90560	4742.97785	6.387223	mutual synchronous 1:1
Ganymede	Jupiter	1.48147862e+23	1.07040000e+09	7.155588	10878.447	7.15455312	synchronous 1:1
Titania	Uranus	3.39960745e+21	4.36298000e+08	8.705869	3644.49384	8.70586546	synchronous 1:1
Oberon	Uranus	3.07597801e+21	5.83511000e+08	13.463237	3151.85103	13.463232	synchronous 1:1
Titan	Saturn	1.34518033e+23	1.22190000e+09	15.945448	5572.69232	15.9454476	synchronous 1:1
Callisto	Jupiter	1.07566088e+23	1.88270000e+09	16.69044	8203.12752	16.689018	synchronous 1:1
Sun		1.98840987e+30				25.3799949	free
Moon	Earth	7.34578925e+22	3.84399000e+08	27.321582	1023.1576	27.3216617	synchronous 1:1
Mercury	Sun	3.30103000e+23	5.79090500e+10	87.9691	47872.0932	58.6461459	spin-orbit 3:2
Iapetus	Saturn	1.80565917e+21	3.56170000e+09	79.331002	3264.97998	79.3308496	synchronous 1:1
Venus	Sun	4.86731000e+24	1.08208000e+11	224.701	35020.3507	243.018484	free retrograde

Appendix C. Phase metric and six-node skeleton

G_phase row	Column 1	Column 2	Column 3
1	2.976410637800	1.243150779500	-2.080360035400
2	1.243150779500	2.592228663500	-2.317883993700
3	-2.080360035400	-2.317883993700	6.773570913300

Object	x	y	z	RA (deg)	Dec (deg)
e1	-0.408953067284	0.884525009431	0.224439070683	114.813083651	+12.969895704
e2	-0.828553895965	-0.256832305478	-0.497529504999	197.222212119	-29.836687522
e3	-0.382434086139	-0.389426083555	0.837909001746	225.519007199	+56.919969623
m_plus	-0.488784996160	-0.101645952212	0.866462536944	191.747587257	+60.050129761
m_minus	0.239773650847	0.640232846849	-0.729801684141	69.468568666	-46.869771113
w_plus	-0.273107431263	0.961095867883	-0.041316627733	105.863163197	-2.367942424
w_minus	-0.505971380457	0.723974494495	0.468885800041	124.948934124	+27.961995685

Appendix D. Source data identity

The report is generated from the following archived source files. SHA-256 hashes identify the exact byte streams used by the computation.

File	Size (bytes)	SHA-256
K-Field_Foundational_Description_and_Derivative_Architecture_1784245726.pdf	3400429	64ccbcaa1ba3fe51e0e7b1e7ce63dcc6ccf27d3620c107b1fcbcd3997b030cf6
K-Field_great_ircle_phase_cost_modulation_1784642570.pdf	1436703	7fd4c9489ba1c8936f5a0b3020b0d6ad513c9fabab1fbb3a47a8e0038319a82
K_Field_recursive_graph_assimilation_100_passes_v1.json	128673	9022232ac9c9581308729557bb33e48385e5b30267ea2d63b764b4a77f9c6024
K_Field_solar_system_rotation_period_quantization_MO_D_1784839005.json	31930	022572c9e74680287b2708a57b15ccb264c0835b99e09e57ec88d2f9bacba2e7
K_Field_solar_system_rotation_period_quantization_MO_D_1784839005_ledger.csv	7255	47fb11e327c15b8fb23216952a3b670e3f836ab259014725ae1a991a793ee7cd
K_Field_solar_system_rotation_period_quantization_MO_D_1784839005_resonances.csv	1120	2d71514d98224bce5a8dd49c8df3a9daee2b401e5e4efbd2ec0e55d328d61b3f
K_Field_solar_system_mass_orbit_kinematics_MOD_1784893627.json	137345	37ea68d27870535553d2670d195f224306adc77c0210df8d1c1c4b5c5d314dbd
K_Field_solar_system_mass_orbit_kinematics_MOD_1784893627_ledger.csv	6195	a215a07b26f0bcebaec5ea4a1650ff71203feaf27ea2035abf67a9a76c5de8cf
K_Field_solar_system_mass_orbit_kinematics_MOD_1784893627_mod_ledger.csv	43583	3099d7d3a2d68046eddb51560699b33c49d6c25e9d12501232484f3f79e77e4c
K_Field_solar_system_mass_orbit_kinematics_MOD_1784893627_log_metric_correlations.csv	1191	26add6904a260f9e6c966c7eb9702d949d7ed678d7513ec3679de8ba5705f89c

Appendix E. Machine-readable companion schema

The companion file K-Field_as_projective_exterior_lattice_hidden_relations_1784897464.json contains the authoritative numerical basis, complete vertex and node ledgers, reciprocal family data, all thirteen shell states, phase eigensystem, solar-system ledgers, PCA, closure residuals, corpus modes, predictions, source hashes, and report metadata. Numeric arrays are stored in explicit SI or nanometre units as labelled. The schema identifier is kfield.projective_exterior_lattice_hidden_relations.v1.

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